

cDFT_solver

Complete User, Input/Output, Workflow, and Developer Manual

Source-audited edition based on the supplied package snapshot

Primary emphasis: rigorous input-file definition, numerical-data formats, output schemas, and explicit end-to-end workflows.

Scope of this edition — This manual was prepared by reading the current Python modules and every bundled Example_workflow input/workflow in the supplied archive. It distinguishes the current code contract from comments or older executor code that no longer matches the active module tree.

Package author identified in the repository README/setup metadata: Vikki Anand Varma.

Manual date: 14 August 2026. Package metadata is internally inconsistent: setup.py reports version 0.2.0, while pyproject.toml reports 0.1.1. This manual therefore identifies the audited source snapshot rather than asserting a single release version.

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Section-level headings are provided within each chapter. Page numbers correspond to this finalized manual.

PART I — Orientation, Installation, and Package Architecture

1. How to Read This Manual

The cDFT_solver package is easiest to understand as a sequence of data transformations. A human-readable .in file is parsed into a nested Python dictionary. Potential modules turn the interaction description into hard-core, mean-field, raw, and combined numerical potential dictionaries. Calculator modules then consume those dictionaries to evaluate correlation functions, free-energy expressions, integrated strengths, thermodynamic states, phase coexistence, or effective potentials.

The key rule — Do not think of the .in file as a loose configuration file. Its hierarchy determines the exact nested dictionary keys later functions search for. A line that is visually similar but produces a different nesting can make a later module silently miss a parameter or fall back to a default.

1.1 The core data pipeline

Stage	Primary object	Produced by	Consumed by
1. Human input	.in text file	User	super_dictionary_creator
2. Parsed configuration	{"system": {...}} dictionary	super_dictionary_creator	all generators/calculators
3. External numerical data	two-column r-value text files	simulation / user	process_supplied_data or potential splitters
4. Potential preprocessing	hc_data, mf_data, raw_data, total_data	potential_splitter modules	free energy, RDF, kernels
5. Symbolic thermodynamics	free_energy result with SymPy variables/expression	total_free_energy	phase_finder / coexistence
6. Structural calculation	g(r), c(r), gamma(r), G_u(r), delta-c, B2	RDF / c_2 / g_alpha modules	analysis and integrated strength
7. State calculation	mu, pressure, free-energy density, vij	phase_finder	EOS tables
8. Coexistence	densities per phase, mu, pressure, vij	coexistence_densities	binodal workflow
9. Inversion	u_ij(r), sigma_ij plus diagnostics	rdf_inversion_advance	effective-interaction analysis

1.2 What is generalized and what is preserved

The worked workflows in Chapters 12–18 preserve the API and physics of the bundled examples but remove dependence on repository folder names. The portable pattern resolves an input path, makes its parent the run directory, and writes generated data to a scratch directory. Exact bundled .in files and their parsed dictionaries are reproduced later so that users can compare their own inputs against a known source snapshot.

2. Installation and Runtime Requirements

The repository README recommends installation directly from GitHub, while `setup.py/pyproject.toml` declare Python ≥ 3.8 and NumPy, SciPy, and Matplotlib. The audited source imports additional packages in active workflows, most importantly SymPy, tqdm, and Numba. The older engine modules also import pyFFTW.

Recommended environment for the demonstrated workflows

```
python -m venv cdft_env
source cdft_env/bin/activate
pip install --upgrade pip
pip install numpy scipy matplotlib sympy tqdm numba
pip install git+https://github.com/vikkivarma16/cDFT_solver.git
```

Compiled OZ library — Several radial/OZ modules load `liboz_radial.so`, `.dylib`, or `.dll` using `ctypes`. The archive includes Linux `.so` files in the relevant subpackages. On another platform or incompatible architecture, the C library may need to be compiled for that system. The C source includes a Linux build hint using LAPACKe, OpenBLAS, and FFTW3.

2.1 Minimal import smoke test

```
from cdft_solver.utils import ExecutionContext, create_unique_scratch_dir
from cdft_solver.generators.parameters.advance_dictionary import super_dictionary_creator
print('cDFT_solver import successful')
```

2.2 Package metadata note

File	Version / requirement observed	Manual interpretation
<code>setup.py</code>	version 0.2.0; Python ≥ 3.8 ; numpy/scipy/matplotlib	Likely newer setup metadata
<code>pyproject.toml</code>	version 0.1.1; Python ≥ 3.8 ; numpy/scipy/matplotlib	Version should be synchronized before release
active calculator imports	sympy, numba; examples use tqdm	Install explicitly for documented workflows
legacy engines	pyfftw plus imports of modules absent from current tree	Do not use as primary workflow in this snapshot

3. Package Architecture

The current package is modular. The Examples/Example_workflow scripts call modules directly, and this direct API is the most reproducible path for the supplied snapshot.

3.1 Directory roles

Directory	Role	Representative modules
<code>cdft_solver/generators/parameters</code>	Parse human input into hierarchical dictionaries	<code>advance_dictionary.py</code>
<code>cdft_solver/generators/potential</code>	Registry and analytic isotropic pair potentials	<code>pair_potential_isotropic_default.py</code> , <code>registry.py</code>

cdft_solver/generators/potential_splitter	Construct hard-core, mean-field, raw, and total potential representations	hc.py, mf.py, raw.py, total.py
cdft_solver/generators/supplied_data	Load file-backed x-y data into numerical arrays	process_supplied_data.py
cdft_solver/calculators/radial_distribution_function	OZ + closure structural calculation	rdf_radial.py, rdf_two_d.py, rdf_planer.py
cdft_solver/calculators/rdf_inversion	Multistate Boltzmann/IBI effective-potential inversion	rdf_inversion_advance.py
cdft_solver/calculators/c_2_analysis	Reference-system, direct-correlation, $\Delta B2$ and $G(r)$ analysis	c_analysis.py
cdft_solver/calculators/g_alpha_r	Coupling-parameter kernels for integrated interaction strength	rdf_alpha_r.py, delta_c.py
cdft_solver/calculators/free_energy_*	Symbolic ideal, mean-field, hard-core and hybrid free energies	ideal.py, mean_field.py, hard_core_bulk_*.py
cdft_solver/calculators/total_free_energy	Merge symbolic components and export them	total_free_energy.py, free_energy_exporter.py
cdft_solver/calculators/integrated_strength	Convert radial kernels to pair strengths v_{ij}	integrated_strength_radial_kernal.py
cdft_solver/calculators/phase_finder	Evaluate one thermodynamic state from density or chemical-potential constraints	thermodynamic_potentials_isocore.py / isochem.py
cdft_solver/calculators/coexistence_densities	Solve multiphase coexistence	calculator_coexistence_densities_isochem.py / isocore.py
cdft_solver/calculators/virial_analysis	Second-virial evaluation and scaling/calibration	second_virial.py, second_virial_scale_calibration.py
cdft_solver/calculators/one_d_profile_iterator	Planar profile iteration support	one_d_profile_iterator_box.py

3.2 Recommended entry points

- For a new calculation, create an ExecutionContext and parse the .in file with super_dictionary_creator.
- For potential/free-energy/EOS work, call hard_core_potentials, meanfield_potentials, raw_potentials, total_potentials, total_free_energy, then a phase-finder routine.
- For structural correlation work, call c_analysis or rdf_radial after parsing the same system dictionary.
- For inversion, first materialize supplied RDF data with process_supplied_data, then call boltzmann_inversion_advance.
- For phase coexistence, build the free-energy result once and then vary constraints on deep copies of the parsed configuration.

Legacy executor status — executor_dft_main.py imports get_unique_dir, which is not present in the current utils.py, and the older engine modules import several generator/calculator modules not present under those names. The direct module workflows bundled in Examples/Example_workflow are therefore documented as the primary interface.

PART II — Input Language, Configuration, and Data Contracts

4. The .in Input Language: Exact Parsing Rules

The parser in `generators/parameters/advance_dictionary.py` reads each nonempty, noncomment line independently. It uses commas to identify multiple attributes or continuation values, then uses the text before the first equals sign to construct hierarchy keys.

4.1 Comments and blank lines

- A line is ignored when it is empty after stripping whitespace.
- A line is ignored when its first non-whitespace character is #.
- A line without = is ignored.
- Inline comments are not stripped by `super_dictionary_creator`. Therefore place comments on separate lines rather than after a value.

4.2 Numeric conversion

Every value is stripped and passed through `float()`. If conversion succeeds, an integral floating value is stored as `int`; otherwise it is stored as `float`. If float conversion fails, the value remains a string. Therefore `yes`, `no`, `None`, `py`, `hnc`, `standard`, and filenames are strings; `1.0` becomes integer `1`; `1e-5` becomes floating `1e-05`.

4.3 Hierarchy construction

```
rdf closure aa = py
solution extrinsic_constraints: density: a = 0.15
interactions primary: aa: type = mie, sigma = 1.0, cutoff = 3.5, epsilon = 0.5, m = 24, n = 12
```

These lines become nested keys by splitting the hierarchy using either colons or whitespace. The final attribute token immediately to the left of the first equals sign is removed from the hierarchy and inserted as an attribute/value at the deepest level.

Resulting nested structure

```
{
  "rdf": {
    "closure": {
      "aa": "py"
    }
  },
  "solution": {
    "extrinsic_constraints": {
      "density": {
        "a": 0.15
      }
    }
  },
  "interactions": {
    "primary": {
      "aa": {
        "type": "mie",
        "sigma": 1,
        "cutoff": 3.5,
        "epsilon": 0.5,
        "m": 24,
        "n": 12
      }
    }
  }
}
```

4.4 Commas: attributes versus list continuation

A comma-separated segment containing = starts a new attribute. A comma-separated segment without = is appended to the previous attribute as another value. This is why `species = a, b` produces `["a", "b"]`, while a one-species line `species = a` produces the string `"a"`.

Recommended species syntax — Use single-character species labels such as `a`, `b`, `c` in the present package and explicitly list all species with commas. Multiple modules create pair keys by concatenating species names and some hard-core code interprets pair keys with `key[0]` and `key[1]`. Multi-character species names are therefore not safe in the current implementation.

4.5 Repeated keys

Repeated dictionary branches are recursively merged. This makes separate lines for `aa`, `ab`, and `bb` interactions or closures natural. Conflicting non-dictionary values can be converted into lists by the generic merge routine, so a user should avoid defining the same scalar key more than once unless a list is intentionally desired.

4.6 base_dict override behavior

When `base_dict` is passed, the parser recursively searches the parsed result for keys present in `base_dict` and replaces the entire matched value. This is powerful but broad: a generic key can match deeper than expected. For reproducible public workflows, explicit edits to the parsed system dictionary are usually clearer than a large `base_dict` override.

5. Master Input-File Reference

This chapter defines the data blocks used by the bundled workflows and the active modules they feed. Not every workflow needs every block.

5.1 species

Input form	Parsed value	Meaning	Downstream assumption
<code>species = a</code>	<code>"a"</code>	one species	Most current modules convert/iterate it successfully because the label has one character.
<code>species = a, b</code>	<code>["a", "b"]</code>	two species	Pair order is <code>aa</code> , <code>ab</code> , <code>bb</code> with symmetric matrix storage.
<code>species = a, b, c</code>	<code>["a", "b", "c"]</code>	three species	Pairs: <code>aa</code> , <code>ab</code> , <code>ac</code> , <code>bb</code> , <code>bc</code> , <code>cc</code> ; every required closure must be provided.

5.2 interactions: primary / secondary / tertiary

Interactions are grouped by level and pair. The raw potential sums all levels for a pair. The mean-field splitter first applies a conversion registry to each level; the hard-core detector inspects every defined level. The current code explicitly recognizes the three level names `primary`, `secondary`, and `tertiary`.

```
interactions primary: aa: type = gs, sigma = 1.414, cutoff = 3.5, epsilon = 2.0
interactions secondary: aa: type = mie, sigma = 1.0, cutoff = 3.5, epsilon = 0.2, m = 12, n = 6
```

A pair may instead be file-backed:

```
interactions primary: aa: filename = pair_aa.txt
```

5.3 Interaction parameter reference

Key	Type	Required?	Meaning in current implementation
type	string	for analytic interaction	Registry key such as gs, lj, mie, hc.
filename	string	alternative to type	Tabulated potential; first column r, second column U(r).
sigma	float	potential dependent	Characteristic length / hard-core diameter; default generally 1.0.
epsilon	float	potential dependent	Energy amplitude; default generally 1.0.
cutoff	float	potential dependent	Finite evaluation range for LJ/Mie/etc.; also influences numerical grid extent.
m	number	Mie/MA/NVRN dependent	In bundled Mie use, larger repulsive exponent, e.g. m=24.
n	number	Mie/GEM/NVRN dependent	Smaller attractive exponent for Mie, or power for generalized Gaussian.
lambda	float	MA	Multiplier of repulsive power-law term.
r_cutoff	float	custom_1	Inner switching radius; defaults to $2^{1/6}\sigma$.
m_i, m_j	float	NVRN	Multiplicative species factors in inverse-power potential.

5.4 Current analytic potential registry

Registry names	Current mathematical behavior	Parameters	Notes
zero / zero_potential	0	none	Identically zero.
hc / ghc / hard_core / hard_sphere	$U=2e8$ for $r \leq \sigma$; 0 outside	sigma	Step-like hard core. Only hc/hardcore set explicit flag=1 in hard-core splitter; ghc provides sigma but does not directly set that flag.
lj / lennard-jones	shifted 12–6 LJ to $U(\text{cutoff})=0$	sigma, epsilon, cutoff	At $r \approx 0$ code uses a large protection value.
gs / gaussian	$\epsilon \exp[-(r/\sigma)^2]$	sigma, epsilon	The callable itself does not truncate at cutoff; grid construction can still use cutoff.

gem / gem_n / ggs / generalized_gaussian	$\epsilon \exp[-(r/\sigma)^n]$	σ, ϵ, n	Generalized Gaussian.
mie	shifted Mie power law	$\sigma, \epsilon, \text{cutoff}, m, n$	Bundled examples use $m > n$, e.g. 24/12.
salj	shifted LJ, flattened below numerical minimum	$\sigma, \epsilon, \text{cutoff}$	Used automatically as mean-field representation of lj.
ma	Mie-like attraction, flattened below numerical minimum	$\sigma, \epsilon, \text{cutoff}, m, n, \lambda$	Used automatically as mean-field representation of mie.
custom_1	piecewise constant/12–6 form	$\sigma, \epsilon, \text{cutoff}, r_{\text{cutoff}}$	Implementation-defined custom potential.
custom_3	9–3-like potential	$\sigma, \epsilon, \text{cutoff}$	Implementation-defined wall-like radial form.
wca	piecewise function in source	$\sigma, \epsilon, \text{cutoff}$	Current code should be interpreted by its implementation; it is not the conventional textbook WCA expression over the full core.
nvrn	shifted inverse-power $\Gamma m_i m_j \sigma^n / r^n$	$\epsilon, m_i, m_j, \sigma, n, \text{cutoff}$	Potential capped at $1e8$.

5.5 Tabulated potential file contract

```
# r    u(r)
0.01   18.4206807
0.02   18.4206807
...
19.99  -1.74e-12
```

Property	Contract / behavior
Columns	At least two numeric columns. Potential splitters use column 1 as r and column 2 as U ; extra columns are not used.
Header	Comment lines beginning with $\#$ are accepted by <code>numpy.loadtxt</code> .
Ordering	Use strictly increasing r for interpolation and reliable grid construction.
Left extrapolation	Tabulated potential interpolation uses the first U value below the file domain.
Right extrapolation	Uses $U=0$ above the file domain.
Path base	Current splitters resolve filename relative to the process current working directory, not automatically relative to <code>ctx.input_file</code> .
Units	No unit conversion is enforced. Use a consistent reduced or physical unit system across r , σ , density, β and potential.

Portable path rule — In a general workflow, resolve the input path and change the working directory to `input_path.parent` before parsing/generating. Then relative filenames in the `.in` file are naturally relative to the

input file location, without depending on any repository folder name.

5.6 rdf block

Input key	Typical value	Consumed by	Meaning / current behavior
rdf closure aa	py / hnc / hybrid	RDF, inversion, c_analysis	Closure for pair aa. Every pair needed by the calculation must be present.
rdf alpha_max	0.01–0.4	OZ iterative solvers	Maximum/adaptive mixing parameter for RDF/OZ iteration.
rdf max_iteration	40000–600000	OZ solvers	Upper iteration limit.
rdf tolerance	1e-5 / 1e-6	many structural modules	OZ convergence tolerance.
rdf tolerence	1e-5 / 1e-6	some supplied examples and some current code paths	Misspelled legacy/example key; see compatibility note below.
rdf beta	1.0	potential scaling, B2, state calculation	Inverse thermal energy beta. No unit conversion.
rdf r_max	5–20	RDF radial grid	Maximum r of structural grid.
rdf n_points	2000	RDF radial grid	Number of interior radial points.
rdf max_iteration_ibi	10000	rdf_inversion_advance	Maximum IBI updates.
rdf alpha_ibi_max	0.1	rdf_inversion_advance	Upper adaptive IBI step factor.
rdf ibi_tolerance	1e-4	rdf_inversion_advance	Convergence threshold on maximum absolute g difference.

Spelling compatibility — The archive contains both tolerance and tolerence. The advanced inversion and c_analysis explicitly read `rdf_block.get("tolerance", 1e-6)`, while the EOS example inputs use tolerence. For new inputs, use tolerance. If a particular function still searches tolerence, its default may be used when tolerance is absent; verify the target module when reproducing an old example.

5.7 Closure names

The built-in registry defines PY, HNC, and hybrid closures. Pair closure matrices are filled symmetrically: if aa, ab, bb are specified for species [a,b], the solver uses ab also for ba.

Name	Role
py	Percus–Yevick closure; especially used for hard/repulsive pairs in the examples.
hnc	Hypernetted-chain closure; used for soft polymer-like pairs in

	the binary examples.
hybrid	Package-defined hybrid closure available in the registry.

5.8 supplied_data: multistate RDF input

```

supplied_data states 1 rdf aa: filename = g_aa.txt
supplied_data states 1 rdf ab: filename = g_ab.txt
supplied_data states 1 rdf bb: filename = g_bb.txt
supplied_data states 1 densities a = 0.10
supplied_data states 1 densities b = 0.02
supplied_data states 1 beta = 1.0

```

`process_supplied_data` recursively replaces each {"filename": ...} leaf with {"x": array, "y": array}. In `rdf_inversion_advance`, each state is then converted to a dense target matrix `g_target` with shape (N,N,Nr) and a Boolean `fixed_mask` with shape (N,N). Missing RDF pairs are left unconstrained rather than automatically invented.

5.9 RDF text-file contract

```

# r    g(r)
0.0175 0.0000000000
0.0525 0.0000000000
...
6.9825 0.9998770921

```

Property	Current behavior
Minimum columns	At least 2. Column 1 becomes x/r; column 2 becomes y/g for the usual inversion case.
Extra columns	<code>process_supplied_data</code> stores all remaining columns as a 2D y array; inversion workflows are safest with exactly one g column per file.
Interpolation below first r	Uses the first supplied g value.
Interpolation above last r	Uses g=1.0.
Tail cleaning	Advanced inversion cleans the interpolated RDF tail before using it as target data.
Density metadata	Each state must provide densities. Missing species in a density dictionary are assigned 0.0 by inversion processing.
Temperature metadata	beta is used if present; otherwise temperature is converted as beta=1/temperature; otherwise beta defaults to 1.0.

5.10 free_energy block

Key	Accepted/current values	Meaning
free_energy mode	standard, hybrid, hybrid_lattice	total_free_energy dispatch. Comments mentioning advanced do not match the current dispatcher.
free_energy method	smf, emf, void	Mean-field symbolic model selected by <code>mean_field.py</code> .
free_energy hs_functional	whitebeer, rosenfeld	Hard-core bulk functional for standard

		mode when sigma is nonzero.
free_energy integrated_strength_kernel	uniform, meanfield, rdf, delta_c	Kernel dispatcher/current v_ij pathway. uniform and meanfield initially create K=1; rdf uses rdf_alpha_r; delta_c uses delta_c_alpha.
free_energy supplied_data	yes/no string in examples	Present in input, but the demonstrated EOS calls pass supplied_data=None to state evaluation.
free_energy ensemble	isochem in binodal example	Used by coexistence configuration.

Important semantic point — In the current EOS source path, integrated_strength_kernel = rdf means “build a coupling-parameter RDF-based kernel internally using rdf_alpha_r,” not “read an RDF text file.” The bundled EOS workflow calls evaluate_canonical_state(... supplied_data=None).

5.11 solution constraints

```
solution extrinsic_constraints: density: a = 0.15
solution extrinsic_constraints: density: b = 0.05
```

Canonical state evaluation requires a density entry for every species. The result uses the species order to build the density vector.

```
solution extrinsic_constraints: number_of_phases = 2
solution extrinsic_constraints: heterogeneous_pair = None
solution extrinsic_constraints: total_density_bound = 10
solution intrinsic_constraints: chemical_potential: a = 14
```

The isochemical coexistence solver treats intrinsic constraints as fixed intensive quantities and solves for phase compositions/densities subject to equality of unconstrained chemical potentials and pressure. In the parsed input, None is the string "None", not Python None; the solver consequently normalizes it as a one-item heterogeneous-pair list, but because its length is not 2 it is not converted to a species pair.

6. ExecutionContext, Paths, and Output Locations

```
from dataclasses import dataclass
from pathlib import Path

@dataclass
class ExecutionContext:
    input_file: Path | None = None
    input_data: str | None = None
    scratch_dir: Path | None = None
    plots_dir: Path | None = None
```

The context is deliberately small. It carries the human input location and the two main output directories. It does not currently contain work_dir, although process_supplied_data checks for that attribute and otherwise falls back to Path.cwd().

6.1 Portable run-directory pattern

```
from pathlib import Path
import os
from cdft_solver.utils import ExecutionContext, create_unique_scratch_dir

input_path = Path('system.in').expanduser().resolve()
os.chdir(input_path.parent) # makes relative data filenames portable
scratch = create_unique_scratch_dir() # created beside the input
plots = scratch / 'plots'
```

```
plots.mkdir(exist_ok=True)

ctx = ExecutionContext(
    input_file=input_path,
    scratch_dir=scratch,
    plots_dir=plots,
)
```

6.2 Typical files written to scratch

Producer	Typical files
super_dictionary_creator	input_system.json
hard_core_potentials	hc.json or supplied_data_potential_hc.json
meanfield_potentials	mf.json or supplied_data_potential_mf.json
raw_potentials	raw.json or supplied_data_potential_raw.json
total_potentials	total.json or supplied_data_potential_total.json
total_free_energy components	fe_id.json, fe_mf.json, fe_hc.json, fe_hybrid.json
free_energy_exporter	fe_total.json
process_supplied_data	supplied_data_materialized.json
c_analysis	result_sigma_analysis.json, delta_B2_results.json, result_G_of_r.json, delta_c_results.json, <prefix>_potential.json
boltzmann_inversion_advance	reference/diagnostic JSON plus <prefix>_potential.json and RDF/potential plots
evaluate_canonical_state	state_from_density.json if export=True

7. Core Output Data Contracts

7.1 Parsed system dictionary

Example parsed output: two-component binodal

```
{
  "system": {
    "species": [
      "a",
      "b"
    ],
    "interactions": {
      "primary": {
        "aa": {
          "type": "gs",
          "sigma": 1,
          "cutoff": 3.5,
          "epsilon": 2
        },
        "ab": {
          "type": "lj",
          "sigma": 1,
          "cutoff": 1.1224,
          "epsilon": 1
        },
        "bb": {
          "type": "mie",
```

```

        "sigma": 1,
        "m": 24,
        "n": 12,
        "cutoff": 2.5,
        "epsilon": 0.1
    }
},
"rdf": {
    "closure": {
        "aa": "py",
        "ab": "py",
        "bb": "py"
    },
    "alpha_max": 0.1,
    "max_iteration": 600000,
    "tolerance": 1e-05,
    "beta": 1,
    "r_max": 10,
    "n_points": 2000
},
"free_energy": {
    "ensemble": "isochem",
    "mode": "standard",
    "method": "smf",
    "hs_functional": "whitebeer",
    "integrated_strength_kernel": "uniform",
    "supplied_data": "no"
},
"solution": {
    "extrinsic_constraints": {
        "number_of_phases": 2,
        "heterogeneous_pair": "None",
        "total_density_bound": 10
    },
    "intrinsic_constraints": {
        "chemical_potential": {
            "a": 14
        }
    }
}
}

```

All later modules use recursive lookup or explicit paths into this structure. A JSON export from `super_dictionary_creator` is therefore an excellent debugging artifact: inspect it before running expensive calculations.

7.2 hard_core_potentials output

```

{
  "species": ["a", "b"],
  "sigma": [[sigma_aa, sigma_ab], [sigma_ba, sigma_bb]],
  "flag": [[flag_aa, flag_ab], [flag_ba, flag_bb]],
  "potentials": {
    "aa": {"r": [...], "U": [...]},
    "ab": {"r": [...], "U": [...]},
    "bb": {"r": [...], "U": [...]}
  }
}

```

`sigma` and `flag` are $N \times N$ symmetric matrices. The generated hard-core potential for each pair is $1e12$ below the effective sigma and zero above it. For tabulated/strongly repulsive interactions, sigma may be estimated using a Barker–Henderson integral after a WCA-like reference split. Off-diagonal diameters are then made additive from self diameters when sufficient information exists.

7.3 meanfield_potentials output

```
{
  "species": ["a", "b"],
  "mf_interactions": {
    "primary": {
      "aa": {...converted interaction...},
      "ab": {...converted interaction...},
      "bb": {...converted interaction...}
    }
  },
  "potentials": {"aa": {"r": [...], "U": [...]}, ...}
}
```

The conversion registry maps hc/ghc/hard_core to zero_potential, lj to salj, and mie to ma before mean-field potential construction. Other registered types are left unchanged.

7.4 raw_potentials output

```
{
  "species": ["a", "b"],
  "potentials": {
    "aa": {"r": [...], "U": [...]},
    "ab": {"r": [...], "U": [...]},
    "bb": {"r": [...], "U": [...]}
  }
}
```

Raw potentials sum all primary/secondary/tertiary contributions without splitting. This is the physical interaction representation used for B2 comparisons and hard-core detection in several workflows.

7.5 total_potentials output

```
{
  "species": ["a", "b"],
  "hc_flag_map": {"aa": 0, "ab": 1, "ba": 1, "bb": 1},
  "total_potentials": {
    "aa": {"r": [...], "U": [...]},
    "ab": {"r": [...], "U": [...]},
    "bb": {"r": [...], "U": [...]}
  }
}
```

The total representation includes the HC potential only for pairs whose hard-core flag is 1; it then adds/interpolates the mean-field representation. Note that many structural-analysis routines deliberately use raw_potentials instead when they need the original full interaction.

7.6 total_free_energy result

```
{
  "variables": (rho_a, rho_b, v_a_a, v_a_b, v_b_a, v_b_b),
  "expression": <SymPy expression>,
  "function": <SymPy Lambda>,
  "components": [...],
  "selected_model": "ideal + mean_field + whitebeer"
}
```

The in-memory result contains SymPy objects. free_energy_exporter serializes symbols and expressions into JSON-safe representations. The phase-finder functions reconstruct symbolic thermodynamics and differentiate the free-energy density with respect to rho_i.

7.7 Canonical state result

```
{
  "ensemble": "canonical",
  "species": ["a", "b"],
  "densities": [rho_a, rho_b],
}
```

```

"vij": [[v_aa, v_ab], [v_ba, v_bb]],
"chemical_potentials": [mu_a, mu_b],
"pressure": P,
"free_energy_density": f
}

```

The thermodynamic relations implemented are $\mu_i = \partial f / \partial \rho_i$ and $P = -f + \sum_i \rho_i \mu_i$. The pair strengths v_{ij} are recalculated for the requested state through the selected integrated-strength kernel.

7.8 Coexistence result

```

{
  "ensemble": "isochem",
  "species": ["a", "b"],
  "n_phases": 2,
  "rhos_per_phase": [[rho_a_1, rho_b_1], [rho_a_2, rho_b_2]],
  "mu_per_phase": [[...], [...]],
  "pressure_per_phase": [P1, P2],
  "vij_per_phase": [[[...]], [...]]
}

```

The solver uses an outer loop because v_{ij} may itself depend on density through the kernel. Within each outer step a nonlinear root solve enforces the specified intrinsic chemical potentials and equality conditions across phases.

7.9 process_supplied_data output

```

{
  "states": {
    "1": {
      "rdf": {
        "aa": {"x": array(...), "y": array(...)},
        "ab": {"x": array(...), "y": array(...)}
      },
      "densities": {"a": 0.10, "b": 0.02},
      "beta": 1
    }
  }
}

```

The returned in-memory object contains NumPy arrays; the optional `supplied_data_materialized.json` export converts those arrays to lists.

7.10 c_analysis principal outputs

Output	Important keys	Physical content
result_sigma_analysis.json	sigma_opt, sigma_bh, bh_meta, r, u_ref, u_real, g/c/gamma for real/reference representations	Reference hard-core mapping and structural comparisons.
delta_B2_results.json	B2_real, B2_ref, delta_B2, 2_delta_B2	Second-virial difference between real and reference potentials.
result_G_of_r.json	r, G_r_sigma_opt, G_r_real, G_u_r_sigma_opt, G_u_r_real, u_attractive_real	Coupling-parameter / thermodynamic-integration kernel diagnostics.
delta_c_results.json	c_real, c_ref_hard, c_sigma_opt, c_rep_sigma_opt, four delta-c arrays	Direct-correlation differences used to construct interaction corrections.
<prefix>_potential.json	pairs -> r, u_r, sigma	Pair potential representation used/exported by the analysis.

8. Physical Quantities and Their Data Flow

8.1 Pair potential $U_{ij}(r)$

The starting interaction is either analytic or tabulated. For each pair, the package may maintain several representations: raw/full U , hard-core reference, and mean-field/soft contribution. Keeping these representations distinct is essential because different physical routes deliberately integrate different forms.

8.2 Hard-core diameter σ_{ij}

The hard-core splitter recognizes explicit hc-like definitions and can infer effective diameters for strongly repulsive analytic or tabulated interactions. The structural/inversion analysis additionally estimates core locations from $g(r)$, performs collective σ optimization, and evaluates Barker–Henderson diameters. These are different definitions and are exported separately rather than silently conflated.

8.3 Radial distribution function $g_{ij}(r)$

The RDF/OZ routines build an $N \times N$ matrix of pair correlation functions on a radial grid $r_k = k \cdot r_{\text{max}} / (n_{\text{points}} + 1)$, $k=1, \dots, n_{\text{points}}$. Closures are supplied pairwise and symmetrized. In inversion, supplied target RDFs are interpolated to this common grid.

8.4 Direct correlation function $c_{ij}(r)$ and $\gamma_{ij}(r)$

The structural solvers maintain $c(r)$ and $\gamma(r)$ alongside $g(r)$. The c_{analysis} workflow compares the direct correlations of real and reference representations and exports negative differences such as $(c_{\text{real}} - c_{\text{ref_hard}})$.

8.5 Second virial coefficient B_2

The implementation uses $B_2 = -2\pi \int [\exp(-\beta U(r)) - 1] r^2 dr$. c_{analysis} exports B_{2_real} , B_{2_ref} , ΔB_2 , and $2 \Delta B_2$ pairwise. The integrated-strength routine also uses $2 \Delta B_2$ for a uniform kernel when a hard core is detected, unless the kernel label requests the direct meanfield route.

8.6 Integrated pair strength v_{ij}

The nominal radial expression is $v_{ij} = \int 4\pi r^2 K_{ij}(r) U_{ij}(r) dr$. Current code contains route-dependent behavior: uniform + hard core can switch to $2 \Delta B_2$; a pure meanfield label directly integrates $K \cdot U_{MF}$; an RDF/coupling kernel supplies a nonuniform radial quantity; δ_{c} supplies a direct-correlation-based kernel.

Source-level caution — In `vij_radial_kernel`, the nonuniform else branch integrates $4\pi r^2 K(r)$ without multiplying by $U(r)$. This matches the current kernel generators, whose returned “value” can already contain the interaction weighting. Do not reinterpret K as a dimensionless $g(r)$ unless you have checked the generator used.

8.7 Free-energy density

Ideal contribution: $f_{id} = \sum_i \rho_i [\ln(\rho_i) - 1]$. Standard mean field: $f_{mf} = (1/2) \sum_{ij} v_{ij} \rho_i \rho_j$. When hard cores are detected and `mode=standard`, a White Bear or Rosenfeld hard-core bulk contribution is added. The symbolic expressions are merged after unifying variables by name.

8.8 Chemical potentials and pressure

From the merged symbolic free-energy density f , the canonical evaluator computes $\mu_i = \partial f / \partial \rho_i$ and $P = -f + \sum_i \rho_i \mu_i$. These are evaluated numerically after v_{ij} is calculated for the current density vector.

8.9 Packing fraction

The binodal example converts density to packing fraction as $\phi_i = \rho_i (\pi/6) d_i^3$ using user-selected effective diameters d_i . These d values are not automatically read from `hc_data` in the supplied example; they are hard-coded post-processing inputs. A general workflow should decide explicitly whether d_i is the hard-core diagonal sigma, a Barker–Henderson diameter, an optimized reference diameter, or another physically defined size.

8.10 Effective potential inversion

The advanced inversion uses one or more target RDF states. If a pair already has a nonzero model potential, that pair is initially protected from inversion; supplied RDF pairs are initialized from $-\ln g$ and marked for inversion. The multistate IBI loop solves OZ for each state, accumulates weighted logarithmic $g_{\text{pred}}/g_{\text{target}}$ corrections, adapts the IBI step factor, and updates the shared pair potential until the maximum RDF deviation meets `ibi_tolerance` or the iteration limit is reached.

9. Input Preflight Checklist

1. Choose single-character species labels and list all species in a deterministic order.
2. For every physical pair required by the solver, provide either an analytic interaction or a tabulated potential file.
3. Provide a closure for every unique pair used by the RDF/OZ calculation.
4. Use `rdf tolerance` for new structural/inversion inputs; document any legacy use of tolerance.
5. Ensure every tabulated potential/RDF file has an increasing first column and at least two numeric columns.
6. Run from the input-file directory or use absolute filenames so current `Path.cwd()`-based loaders resolve files correctly.
7. Parse the input and inspect `input_system.json` before expensive calculation.
8. Check the hard-core sigma and flag matrices before accepting a free-energy model.
9. For canonical calculations, provide density for every species.
10. For inversion, provide density and beta/temperature metadata for every supplied state.
11. For coexistence, confirm the intrinsic-constraint count does not exceed $N_{\text{species}}-1$ and set a physically meaningful `total_density_bound`.
12. Keep beta, length, potential and density units mutually consistent; the package does not perform dimensional-unit conversion.

PART III — Reusable Workflows and Physical Calculations

10. General Reusable Workflow Skeleton

Portable skeleton

```
from pathlib import Path
import os

from cdft_solver.utils import ExecutionContext, create_unique_scratch_dir
```

```

from cdft_solver.generators.parameters.advance_dictionary import super_dictionary_creator

# 1) Locate the input without assuming a repository folder structure
input_path = Path('system.in').expanduser().resolve()
os.chdir(input_path.parent)

# 2) Create run outputs next to the chosen input
scratch = create_unique_scratch_dir()
plots = scratch / 'plots'
plots.mkdir(exist_ok=True)

# 3) Build the execution context
ctx = ExecutionContext(
    input_file=input_path,
    scratch_dir=scratch,
    plots_dir=plots,
)

# 4) Parse and export the exact machine-readable configuration
system = super_dictionary_creator(
    ctx=ctx,
    export_json=True,
    filename='input_system.json',
    super_key_name='system',
)

print(system)
print('Run directory:', scratch)

```

All later workflow examples in this manual start from this pattern. Only the calculation-specific modules and parameter loops change.

11. Potential and Free-Energy Preprocessing Workflow

Reusable preprocessing block

```

from cdft_solver.generators.potential_splitter.hc import hard_core_potentials
from cdft_solver.generators.potential_splitter.mf import meanfield_potentials
from cdft_solver.generators.potential_splitter.raw import raw_potentials
from cdft_solver.generators.potential_splitter.total import total_potentials
from cdft_solver.calculators.total_free_energy.total_free_energy import total_free_energy
from cdft_solver.calculators.total_free_energy.free_energy_exporter import free_energy_exporter

hc_data = hard_core_potentials(
    ctx=ctx, input_data=system, grid_points=5000,
    file_name_prefix='hc.json', export_files=True,
)

mf_data = meanfield_potentials(
    ctx=ctx, input_data=system, grid_points=5000,
    file_name_prefix='mf.json', export_files=True,
)

raw_data = raw_potentials(
    ctx=ctx, input_data=system, grid_points=5000,
    file_name_prefix='raw.json', export_files=True,
)

total_data = total_potentials(
    ctx=ctx, hc_source=hc_data, mf_source=mf_data,
    file_name_prefix='total.json', export_files=True,
)

filenames = {
    'hard_core': 'fe_hc.json',
    'mean_field': 'fe_mf.json',
    'ideal': 'fe_id.json',
}

```

```

    'hybrid': 'fe_hybrid.json',
}

free_energy = total_free_energy(
    ctx=ctx,
    hc_data=hc_data,
    system_config=system,
    export_json=True,
    filenames=filenames,
)

free_energy_exporter(
    ctx=ctx,
    total_fe=free_energy,
    filename='fe_total.json',
    indent=4,
)

```

11.1 What to inspect before proceeding

Object	Check	Reason
system	species, interactions, rdf/free_energy/solution nesting	Confirms parser interpreted the .in file as intended.
hc_data["sigma"]	size and magnitude of effective diameters	Hard-core free energy and reference systems depend on this.
hc_data["flag"]	which pairs are treated as hard core	Controls total potential and hard-core inclusion.
mf_data["mf_interactions"]	automatic lj→salj, mie→ma, hc→zero conversions	Confirms perturbative representation.
raw_data	full pair interaction curves	Used by B2 and structural reference analysis.
free_energy["selected_model"]	actual assembled model	Confirms mode/functional dispatch.
free_energy variables	density and v _{ij} symbol order	Phase evaluation flattens values into this symbolic order.

12. Worked Workflow A — One-Component Equation of State

Goal: evaluate chemical potential, pressure, free-energy density, and integrated pair strength over a density grid for one species.

12.1 General input file

```

species = a

interactions primary: aa: filename = pair_aa.txt

rdf closure aa = py
rdf alpha_max = 0.01
rdf r_max = 5.0
rdf max_iteration = 200000
rdf tolerance = 0.000001
rdf n_points = 2000
rdf beta = 1.0

free_energy mode = standard
free_energy hs_functional = whitebeer

```

```

free_energy method = smf
free_energy integrated_strength_kernel = rdf
free_energy supplied_data = no

solution extrinsic_constraints: density: a = 0.15

```

12.2 Field-by-field interpretation

Line/block	Why it exists	What later code reads
species = a	Defines N=1 and pair aa.	species lookup; density variable rho_a; v_a_a.
aa filename	Provides full physical pair potential.	HC/MF/raw splitters use it from current working directory.
rdf closure aa	Defines OZ closure.	rdf_alpha_r when kernel=rdf.
rdf grid/iteration values	Controls the internally calculated structural kernel.	rdf_alpha_r and OZ solver.
free_energy mode/method/hs_functional	Chooses symbolic free-energy model.	total_free_energy and mean_field.
integrated_strength_kernel=rdf	Makes v_aa density dependent through internally calculated coupling-route kernel.	build_strength_kernel -> rdf_alpha_r -> vij_radial_kernel.
density a=0.15	Provides a valid initial canonical density in the parsed config.	Workflow overwrites it for every EOS point.

12.3 Generalized EOS script

Portable one-component EOS

```

import json
import numpy as np

from cdft_solver.calculators.phase_finder.thermodynamic_potentials_isocore import evaluate_canonical_state

# Assume ctx, system and free_energy were created by Chapters 10–11.
rho_values = np.linspace(0.05, 0.80, 30)
results = []

for rho in rho_values:
    system['system']['solution']['extrinsic_constraints']['density']['a'] = float(rho)

    state = evaluate_canonical_state(
        ctx=ctx,
        config_dict=system,
        fe_res=free_energy,
        supplied_data=None,
        export=False,
        verbose=False,
    )

    results.append({
        'rho': float(rho),
        'v_aa': state['vij'][0][0],
        'mu_a': state['chemical_potentials'][0],
        'pressure': state['pressure'],
        'free_energy_density': state['free_energy_density'],
    })

output = scratch / 'eos_one_component.json'

```

```
output.write_text(json.dumps(results, indent=2))
```

12.4 Data interpretation

Each EOS point is a complete canonical state. If the kernel is density dependent, v_{aa} changes with ρ ; therefore it should be stored alongside pressure and chemical potential rather than treated as one global constant. If the kernel is uniform/meanfield, v_{aa} may become density independent for a fixed potential, depending on the hard-core/B2 route selected by `vij_radial_kernel`.

12.5 Bundled example specifics

The supplied one-component EOS workflow uses a manually specified density array rather than `linspace`, updates density `a` inside the parsed dictionary, calls `evaluate_canonical_state`, and writes `rho`, `vij`, `mu` and pressure. Its input uses the legacy spelling `rdf` tolerance and refers to a tabulated `aa` potential. The generalized version above retains the same physics but uses `rdf` tolerance and writes into `scratch`.

13. Worked Workflow B — Two-Component Equation of State at Fixed Composition

Goal: scan total density while holding mole fraction x_b fixed. For every total density ρ_t , the workflow sets $\rho_a = \rho_t(1 - x_b)$ and $\rho_b = \rho_t x_b$, then performs the same canonical thermodynamic evaluation.

13.1 General input

```
species = a, b

interactions primary: aa: filename = pair_aa.txt
interactions primary: ab: filename = pair_ab.txt
interactions primary: bb: filename = pair_bb.txt

rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.01
rdf r_max = 3.5
rdf max_iteration = 200000
rdf tolerance = 0.000001
rdf n_points = 2000
rdf beta = 1.0

free_energy mode = standard
free_energy hs_functional = whitebeer
free_energy method = smf
free_energy integrated_strength_kernel = rdf
free_energy supplied_data = no

solution extrinsic_constraints: density: a = 0.15
solution extrinsic_constraints: density: b = 0.0
```

13.2 Generalized script

Portable binary EOS

```
import json
import numpy as np
from cdft_solver.calculators.phase_finder.thermodynamic_potentials_isocore import evaluate_canonical_state

rho_total_values = np.linspace(0.05, 0.90, 35)
x_b = 0.167
results = []

for rho_total in rho_total_values:
    rho_a = rho_total * (1.0 - x_b)
```

```

rho_b = rho_total * x_b

system['system']['solution']['extrinsic_constraints']['density']['a'] = float(rho_a)
system['system']['solution']['extrinsic_constraints']['density']['b'] = float(rho_b)

state = evaluate_canonical_state(
    ctx=ctx, config_dict=system, fe_res=free_energy,
    supplied_data=None, export=False, verbose=False,
)

results.append({
    'rho_total': float(rho_total),
    'rho_a': float(rho_a),
    'rho_b': float(rho_b),
    'x_a': float(1.0-x_b),
    'x_b': float(x_b),
    'vij': state['vij'],
    'mu_a': state['chemical_potentials'][0],
    'mu_b': state['chemical_potentials'][1],
    'pressure': state['pressure'],
    'free_energy_density': state['free_energy_density'],
})

(scratch/'eos_binary.json').write_text(json.dumps(results, indent=2))

```

13.3 Ordering contract

For species [a,b], densities and chemical potentials are ordered [a,b]. v_{ij} is a 2×2 matrix ordered [[aa,ab], [ba,bb]]. The symbolic mean-field free energy contains both v_{ab} and v_{ba} symbols, although numerically the pair-strength calculation fills them symmetrically.

13.4 Difference between supplied 12–6 and 24–12 examples

The two repository EOS subfolders use the same Python workflow and input structure. Their difference is in the tabulated aa/ab/bb potential files. A general user should therefore think of “12–6” or “24–12” as data choices, not as workflow types or required directory names.

14. Worked Workflow C — One-Component Effective-Potential Inversion from RDF

Goal: infer an effective pair potential from a target $g_{aa}(r)$ measured at a known density and beta.

14.1 General inversion input

```

species = a

rdf closure aa = py
rdf alpha_max = 0.1
rdf max_iteration = 40000
rdf max_iteration_ibi = 10000
rdf alpha_ibi_max = 0.1
rdf tolerance = 0.00001
rdf ibi_tolerance = 0.0001
rdf beta = 1.0
rdf r_max = 20.0
rdf n_points = 2000

supplied_data states 1 rdf aa: filename = g_aa.txt
supplied_data states 1 densities a = 0.4
supplied_data states 1 beta = 1.0

```

14.2 Why there is no interactions block

The bundled one-component inversion example comments out the model interactions. With no interactions block, the hard-core/mean-field preprocessors initially construct zero potentials. The

supplied g_aa is therefore marked for inversion, initialized approximately by $-\ln g$, and updated by IBI. This is appropriate when the effective potential itself is the unknown.

14.3 Generalized script

Portable one-component inversion

```
from cdft_solver.generators.supplied_data.process_supplied_data import process_supplied_data
from cdft_solver.calculators.rdf_inversion.rdf_inversion_advance import boltzmann_inversion_advance

supplied = process_supplied_data(
    ctx=ctx,
    config=system,
    export_json=True,
    export_plot=True,
)

u_matrix, sigma_matrix = boltzmann_inversion_advance(
    ctx=ctx,
    rdf_config=system,
    supplied_data=supplied,
    filename_prefix='effective',
    export_plot=True,
    export_json=True,
)

print('u_matrix shape:', u_matrix.shape)      # (N,N,Nr)
print('sigma_matrix shape:', sigma_matrix.shape) # (N,N)
```

14.4 IBI data flow

13. process_supplied_data loads g_aa.txt and replaces filename with x and y arrays.
14. rdf_inversion_advance interpolates the target RDF to its internal r grid.
15. The target pair is marked in fixed_mask and initialized from $-\ln(\max(g, g_{\text{floor}}))$.
16. For each IBI iteration, the OZ solver predicts g_pred from the current u(r).
17. The update accumulates $(\beta_{\text{ref}}/\beta_{\text{state}}) \cdot \ln(g_{\text{pred}}/g_{\text{target}})$ on valid r values.
18. An adaptive alpha_ibi multiplies the update and is bounded between a small minimum and alpha_ibi_max.
19. When $\max|g_{\text{pred}} - g_{\text{target}}| < \text{ibi_tolerance}$, the code stores final c, gamma and g for later reference/sigma analysis.
20. The routine returns u_matrix/beta_ref and sigma_matrix and writes pairwise JSON/plots when enabled.

15. Worked Workflow D — Two-Component Multistate/Multipair Potential Inversion

Goal: infer shared aa, ab and bb effective pair potentials from target RDFs at a known binary state. The same input structure can be extended to multiple states by adding states 2, states 3, etc.

15.1 General input

```
species = a, b

rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.1
rdf max_iteration = 40000
rdf max_iteration_ibi = 10000
rdf alpha_ibi_max = 0.1
```

```

rdf tolerance = 0.00001
rdf ibi_tolerance = 0.0001
rdf beta = 1.0
rdf r_max = 20.0
rdf n_points = 2000

supplied_data states 1 rdf aa: filename = g_aa.txt
supplied_data states 1 rdf ab: filename = g_ab.txt
supplied_data states 1 rdf bb: filename = g_bb.txt
supplied_data states 1 densities a = 0.109860935
supplied_data states 1 densities b = 0.022041065
supplied_data states 1 beta = 1.0

```

15.2 Extending to two thermodynamic states

```

supplied_data states 2 rdf aa: filename = g_aa_state2.txt
supplied_data states 2 rdf ab: filename = g_ab_state2.txt
supplied_data states 2 rdf bb: filename = g_bb_state2.txt
supplied_data states 2 densities a = 0.15
supplied_data states 2 densities b = 0.03
supplied_data states 2 beta = 1.0

```

The inversion assigns uniform weights $1/N_{\text{states}}$ to the states. A single shared $u_{ij}(r)$ is updated from the weighted sum of state corrections, which is the important distinction from running independent inversions state by state.

15.3 Output interpretation

u_{matrix} has shape (2,2,Nr) and is symmetric. σ_{matrix} has shape (2,2). The exported JSON uses pair keys such as aa, ab, ba, bb and stores r , u_r and σ . Additional diagnostic files compare real/reference structures, evaluate coupling quantities, and can export RDF comparison plots for each supplied state and pair.

16. Worked Workflow E — One-Component Structural and Direct-Correlation Analysis

Goal: start from a known pair potential at a selected density and generate real/reference $g(r)$, $c(r)$, $\gamma(r)$, effective core diameters, Delta B2, coupling-route G quantities, and Delta c diagnostics.

16.1 General input

```

species = a
interactions primary: aa: filename = pair_aa.txt

rdf closure aa = py
rdf alpha_max = 0.4
rdf max_iteration = 400000
rdf tolerance = 0.000001
rdf beta = 1.0
rdf r_max = 20.0
rdf n_points = 2000

```

16.2 Generalized script

Portable structural analysis

```

import numpy as np
from cdft_solver.calculators.c_2_analysis.c_analysis import c_analysis

rho = np.array([0.4], dtype=float)
u_matrix, sigma_matrix = c_analysis(
    ctx=ctx,
    rdf_config=system,
    supplied_data=None,

```

```

densities=rho,
filename_prefix='structure',
export_plot=True,
export_json=True,
)

```

16.3 Explicit stages performed inside c_analysis

21. Generate HC, MF, total and raw representations from the same system configuration.
22. Build the RDF radial grid from rdf r_max and n_points.
23. Construct the closure matrix from pairwise closure entries.
24. Interpolate the raw physical potential onto the RDF grid and solve OZ for the full system.
25. Detect hard-core locations from g(r), construct reference potentials, and optimize reference sigma where cores are present.
26. Compute Barker-Henderson sigma values for comparison.
27. Solve g/c/gamma for real and reference systems and export result_sigma_analysis.json.
28. Compute pairwise B2_real, B2_ref, Delta B2 and 2 Delta B2.
29. Build attractive-tail representation and coupling-parameter G(r) / G_u(r) quantities.
30. Construct c(r) differences among real/reference/optimized representations and export delta_c_results.json.
31. Export potential JSON and figures.

17. Worked Workflow F — Two-Component Structural Correlation Analysis

The binary version is structurally identical to Chapter 16, but every scalar pair quantity becomes a symmetric 2x2 matrix and each radial quantity becomes an array with shape (2,2,Nr).

17.1 General input

```

species = a, b

interactions primary: aa: filename = pair_aa.txt
interactions primary: ab: filename = pair_ab.txt
interactions primary: bb: filename = pair_bb.txt

rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.1
rdf max_iteration = 400000
rdf tolerance = 0.00001
rdf beta = 1.0
rdf r_max = 10.0
rdf n_points = 2000

```

17.2 Generalized script

```

import numpy as np
from cdft_solver.calculators.c_2_analysis.c_analysis import c_analysis

densities = np.array([0.60, 0.096], dtype=float)

u_matrix, sigma_matrix = c_analysis(
    ctx=ctx,
    rdf_config=system,
    supplied_data=None,
    densities=densities,
    filename_prefix='binary_structure',
    export_plot=True,
    export_json=True,
)

```

17.3 Pair-array reading guide

Quantity	Shape for N species	Binary interpretation
r	(Nr,)	common radial grid
u_matrix	(N,N,Nr)	u[0,0]=aa; u[0,1]=ab; u[1,1]=bb
g/c/gamma	(N,N,Nr)	same pair indexing
sigma_opt / sigma_bh	(N,N)	effective pair diameters
B2 matrices	(N,N)	pair virial coefficients
delta_c arrays	(N,N,Nr)	pair direct-correlation corrections

18. Worked Workflow G — Two-Component Isochemical Binodal

Goal: determine two-phase coexistence while fixing one species chemical potential and allowing the other species to satisfy equality of chemical potentials and pressure between phases.

18.1 General input

```
species = a, b

interactions primary: aa: type = gs, sigma = 1.0, cutoff = 3.5, epsilon = 2.0
interactions primary: ab: type = lj, sigma = 1.0, cutoff = 1.1224, epsilon = 1.0
interactions primary: bb: type = mie, sigma = 1.0, m = 24, n = 12, cutoff = 2.5, epsilon = 0.1

rdf closure aa = py
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.1
rdf max_iteration = 600000
rdf tolerance = 0.00001
rdf beta = 1.0
rdf r_max = 10.0
rdf n_points = 2000

free_energy ensemble = isochem
free_energy mode = standard
free_energy method = smf
free_energy hs_functional = whitebeer
free_energy integrated_strength_kernel = uniform
free_energy supplied_data = no

solution extrinsic_constraints: number_of_phases = 2
solution extrinsic_constraints: heterogeneous_pair = None
solution extrinsic_constraints: total_density_bound = 10
solution intrinsic_constraints: chemical_potential: a = 14
```

18.2 Why the free energy is built once

The symbolic free-energy form depends on the selected model and effective hard-core data, not on a specific trial density. Therefore the workflow builds `hc_data` and `free_energy` once, then changes the chemical-potential and density-bound constraints on a deep copy of the system before each coexistence call. Density-dependent v_{ij} values are updated inside the coexistence outer loop.

18.3 Generalized binodal scan

Portable coexistence scan

```
import copy
import numpy as np

from cdft_solver.calculators.coexistence_densities.calculator_coexistence_densities_isochem import
coexistence_densities_isochem

mu_a_values = np.linspace(7.0, 10.0, 10)
density_bounds = np.linspace(5.0, 10.0, len(mu_a_values))
rows = []

for mu_a, rho_bound in zip(mu_a_values, density_bounds):
    cfg = copy.deepcopy(system)
    cfg['system']['solution']['intrinsic_constraints']['chemical_potential']['a'] = float(mu_a)
    cfg['system']['solution']['extrinsic_constraints']['total_density_bound'] = float(rho_bound)

    sol = coexistence_densities_isochem(
        ctx=ctx,
        config_dict=cfg,
        fe_res=free_energy,
        supplied_data=None,
        max_outer_iters=10,
        tol_outer=1e-3,
        tol_solver=1e-8,
        verbose=False,
    )

    if sol is None:
        continue

    phase1, phase2 = np.asarray(sol['rhos_per_phase'], dtype=float)
    if np.allclose(phase1, phase2, atol=1e-6):
        continue

    rows.append({
        'mu_a': float(mu_a),
        'rho_phase1': phase1.tolist(),
        'rho_phase2': phase2.tolist(),
        'pressure_phase1': sol['pressure_per_phase'][0],
        'pressure_phase2': sol['pressure_per_phase'][1],
        'vij_per_phase': sol['vij_per_phase'],
    })
```

18.4 Converting to packing fraction

```
from math import pi

def packing_fraction(rho, diameter):
    return rho * (pi/6.0) * diameter**3

# Decide the physical diameter definition explicitly.
d_a = hc_data['sigma'][0][0]
d_b = hc_data['sigma'][1][1]

phi_a_1 = packing_fraction(rows[0]['rho_phase1'][0], d_a)
phi_b_1 = packing_fraction(rows[0]['rho_phase1'][1], d_b)
```

Difference from bundled example — The bundled workflow hard-codes `d_a` and `d_b` for packing-fraction post-processing. The generalized example above uses the diagonal HC sigma only as a transparent default. If your scientific definition requires optimized or Barker–Henderson diameters, use those instead.

18.5 Solver constraints

For N species, the isochemical solver rejects more than N-1 independent intrinsic constraints. It parameterizes each phase by total density and N-1 sequential fractions, generates random initial guesses, solves the coexistence residual with `scipy.optimize.root`, rejects nearly identical phases, applies density bounds, and iterates v_{ij} across outer iterations until the density change norm is below `tol_outer` or the outer iteration limit is reached.

19. Physical-Quantity Cookbook

19.1 Compute a radial RDF directly

```
import numpy as np
from cdft_solver.calculators.radial_distribution_function.rdf_radial import rdf_radial

densities = np.array([0.2, 0.05])
rdf = rdf_radial(
    ctx=ctx,
    rdf_config=system,
    densities=densities,
    supplied_data=None,
    export=True,
    plot=True,
    filename_prefix='rdf_state',
)
```

Use the same system input: interactions define U_{ij} , rdf closure/grid keys define the OZ problem, and densities are passed explicitly as a NumPy array in species order.

19.2 Compute second virial data

The package also exposes `calculators/virial_analysis/second_virial.py` and `second_virial_scale_calibration.py`. These are not used by the bundled `Example_workflow` scripts, so treat them as advanced API. `c_analysis` provides the most explicit demonstrated B2 output by comparing real and reference pair potentials.

19.3 Evaluate one canonical state without an EOS loop

```
state = evaluate_canonical_state(
    ctx=ctx,
    config_dict=system,
    fe_res=free_energy,
    supplied_data=None,
    export=True,
    output_file='state.json',
    verbose=True,
)
```

19.4 Change composition safely

Always modify the nested density dictionary by species name rather than relying on positional assumptions. The evaluator itself rebuilds the density vector in species order. This makes the script readable and prevents accidental a/b reversal.

19.5 Reuse a parsed base configuration

For parameter scans, use `copy.deepcopy(system)` before changing chemical potentials, bounds, or other constraints when independent runs must not modify a shared base object. For a simple EOS loop where only one current state exists at a time, mutating the density entries in place is sufficient.

PART IV — Input Design, Validation, and Audited Examples

20. Troubleshooting and Current-Implementation Notes

Symptom	Likely cause / action
FileNotFoundError for pair potential	Potential splitters use <code>os.getcwd()</code> with the filename. Run from the input/data directory, use an absolute filename, or adopt the portable <code>os.chdir(input_path.parent)</code> pattern.
Supplied RDF file not found	<code>process_supplied_data</code> also falls back to <code>Path.cwd()</code> because <code>ExecutionContext</code> has no <code>work_dir</code> field.
NameError: SuppliedDataError	<code>process_supplied_data</code> raises <code>SuppliedDataError</code> but the class is not defined/imported in that module. Valid inputs avoid the branch; malformed/missing supplied data can expose this source bug.
Unknown free energy mode advanced	Current <code>total_free_energy</code> accepts standard, hybrid, hybrid_lattice. Old comments mentioning advanced are stale.
Unknown method zdl	Current <code>mean_field</code> dispatch accepts smf, emf, void. Old comments calling the third method zdl do not match the current key.
tolerance seems ignored	Check whether the target module reads tolerance or the legacy misspelling tolerance.
Single species behaves strangely	Use a one-character label. Parser returns "a" rather than ["a"], but many modules rely on iterating characters and therefore only one-character scalar labels are safe.
Pair names with long species labels fail	Several modules split pair keys as <code>key[0]</code> , <code>key[1]</code> . Current safe convention is a, b, c.
Unexpected "None" heterogeneous pair	The parser stores None as the string "None". Prefer omitting the line when no heterogeneous pair constraint is needed, or be aware the current solver ignores it as a pair because its string length is not 2.
EOS is very slow	<code>rdf</code> integrated strength calls an internal structural/coupling calculation for each state. uniform/meanfield kernels are cheaper but represent a different physical approximation.
OZ shared-library load error	The platform-specific <code>liboz_radial</code> binary may not match the OS/architecture; compile the accompanying C source and required numerical libraries.
Install succeeds but workflow import fails	Declared dependencies omit some active imports such as <code>sympy</code> , <code>numba</code> and <code>tqdm</code> . Install them explicitly.
executor_dft_main import error	The legacy executor references <code>get_unique_dir</code> and engine dependencies absent from the current module tree. Use direct module workflows.
Potential is not zero at stated Gaussian cutoff	The Gaussian/GEM callables do not enforce cutoff internally; cutoff can still influence grid extent. If strict truncation is required, provide a tabulated truncated potential or modify the potential implementation.
Mie exponent confusion	Follow the supplied convention <code>m</code> =larger repulsive exponent and <code>n</code> =smaller exponent, e.g. <code>m=24,n=12</code> .

21. Designing a New Input from Scratch

21.1 One-component soft fluid

```
species = a
interactions primary: aa: type = gs, sigma = 1.0, cutoff = 5.0, epsilon = 2.0
rdf closure aa = hnc
rdf alpha_max = 0.1
rdf max_iteration = 100000
rdf tolerance = 1e-6
rdf beta = 1.0
rdf r_max = 10.0
rdf n_points = 2000
free_energy mode = standard
free_energy method = smf
free_energy hs_functional = whitebeer
free_energy integrated_strength_kernel = meanfield
free_energy supplied_data = no
solution extrinsic_constraints: density: a = 0.2
```

21.2 Binary soft/hard mixture

```
species = a, b
interactions primary: aa: type = gs, sigma = 1.414, cutoff = 5.0, epsilon = 2.0
interactions primary: ab: type = lj, sigma = 1.0, cutoff = 1.122462, epsilon = 1.0
interactions primary: bb: type = hc, sigma = 1.0
rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.05
rdf max_iteration = 200000
rdf tolerance = 1e-6
rdf beta = 1.0
rdf r_max = 10.0
rdf n_points = 2000
free_energy mode = standard
free_energy method = smf
free_energy hs_functional = whitebeer
free_energy integrated_strength_kernel = uniform
free_energy supplied_data = no
solution extrinsic_constraints: density: a = 0.2
solution extrinsic_constraints: density: b = 0.05
```

21.3 Three-species closure completeness

For species a,b,c, provide closures for aa, ab, ac, bb, bc, cc. Pair potentials should also cover all physically nonzero pairs. If a pair is truly zero, define type=zero explicitly or verify the downstream generator's no-interactions fallback is appropriate for the whole system.

22. Audited Bundled Input Files and Exact Parsed Dictionaries

This chapter reproduces the active, uncommented input content in the supplied examples and the exact nested dictionary created by `super_dictionary_creator`. It is intended as a debugging reference.

One-component EOS

Active input lines

```
species = a
interactions primary: aa: filename = aa_colloids.txt
rdf closure aa = py
rdf alpha_max = 0.01
rdf r_max = 5.0
rdf max_iteration = 200000
rdf tolerance = 0.000001
```

```

rdf n_points = 2000
rdf beta = 1.0
free_energy mode = standard
free_energy hs_functional = whitebeer
free_energy method = smf
free_energy integrated_strength_kernel = rdf
free_energy supplied_data = no
solution extrinsic_constraints: density: a = 0.15

```

Exact parsed dictionary

```

{
  "system": {
    "species": "a",
    "interactions": {
      "primary": {
        "aa": {
          "filename": "aa_colloids.txt"
        }
      }
    }
  },
  "rdf": {
    "closure": {
      "aa": "py"
    },
    "alpha_max": 0.01,
    "r_max": 5,
    "max_iteration": 200000,
    "tolerance": 1e-06,
    "n_points": 2000,
    "beta": 1
  },
  "free_energy": {
    "mode": "standard",
    "hs_functional": "whitebeer",
    "method": "smf",
    "integrated_strength_kernel": "rdf",
    "supplied_data": "no"
  },
  "solution": {
    "extrinsic_constraints": {
      "density": {
        "a": 0.15
      }
    }
  }
}

```

One-component potential inversion

Active input lines

```

species = a
rdf closure aa = py
rdf alpha_max = 0.1
rdf max_iteration = 40000
rdf max_iteration_ibi = 10000
rdf alpha_ibi_max = 0.1
rdf tolerance = 0.00001
rdf ibi_tolerance = 0.0001
rdf beta = 1.0
rdf r_max = 20.0
rdf n_points = 2000
supplied_data states 1 rdf aa: filename = rdf_colloid.txt
supplied_data states 1 densities a = 0.4
supplied_data states 1 beta = 1.0

```

Exact parsed dictionary

```
{
  "system": {
    "species": "a",
    "rdf": {
      "closure": {
        "aa": "py"
      },
      "alpha_max": 0.1,
      "max_iteration": 40000,
      "max_iteration_ibi": 10000,
      "alpha_ibi_max": 0.1,
      "tolerance": 1e-05,
      "ibi_tolerance": 0.0001,
      "beta": 1,
      "r_max": 20,
      "n_points": 2000
    },
    "supplied_data": {
      "states": {
        "1": {
          "rdf": {
            "aa": {
              "filename": "rdf_colloid.txt"
            }
          },
          "densities": {
            "a": 0.4
          },
          "beta": 1
        }
      }
    }
  }
}
```

One-component structural analysis**Active input lines**

```
species = a
interactions primary: aa: filename = aa_colloids.txt
rdf closure aa = py
rdf alpha_max = 0.4
rdf max_iteration = 400000
rdf tolerance = 0.000001
rdf beta = 1.0
rdf r_max = 20.0
rdf n_points = 2000
```

Exact parsed dictionary

```
{
  "system": {
    "species": "a",
    "interactions": {
      "primary": {
        "aa": {
          "filename": "aa_colloids.txt"
        }
      }
    },
    "rdf": {
      "closure": {
        "aa": "py"
      },
      "alpha_max": 0.4,
      "max_iteration": 400000,
```

```

    "tolerance": 1e-06,
    "beta": 1,
    "r_max": 20,
    "n_points": 2000
  }
}
}

```

Two-component potential inversion

Active input lines

```

species = a, b
rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.1
rdf max_iteration = 40000
rdf max_iteration_ibi = 10000
rdf alpha_ibi_max = 0.1
rdf tolerance = 0.00001
rdf ibi_tolerance = 0.0001
rdf beta = 1.0
rdf r_max = 20.0
rdf n_points = 2000
supplied_data states 1 rdf aa: filename = average_g11.txt
supplied_data states 1 rdf ab: filename = average_g12.txt
supplied_data states 1 rdf bb: filename = average_g22.txt
supplied_data states 1 densities a = 0.109860935
supplied_data states 1 densities b = 0.022041065
supplied_data states 1 beta = 1.0

```

Exact parsed dictionary

```

{
  "system": {
    "species": [
      "a",
      "b"
    ],
    "rdf": {
      "closure": {
        "aa": "hnc",
        "ab": "py",
        "bb": "py"
      },
      "alpha_max": 0.1,
      "max_iteration": 40000,
      "max_iteration_ibi": 10000,
      "alpha_ibi_max": 0.1,
      "tolerance": 1e-05,
      "ibi_tolerance": 0.0001,
      "beta": 1,
      "r_max": 20,
      "n_points": 2000
    },
    "supplied_data": {
      "states": {
        "1": {
          "rdf": {
            "aa": {
              "filename": "average_g11.txt"
            },
            "ab": {
              "filename": "average_g12.txt"
            },
            "bb": {
              "filename": "average_g22.txt"
            }
          }
        }
      }
    }
  }
}

```

```
    },  
    "densities": {  
      "a": 0.109860935,  
      "b": 0.022041065  
    },  
    "beta": 1  
  }  
}  
}
```

Two-component EOS (12-6 case)

Active input lines

```
species = a, b
interactions primary: aa: filename = aa.txt
interactions primary: ab: filename = ab.txt
interactions primary: bb: filename = bb.txt
rdf closure aa = hnc
rdf closure bb = py
rdf closure ab = py
rdf alpha_max = 0.01
rdf r_max = 3.5
rdf max_iteration = 200000
rdf tolerance = 0.000001
rdf n_points = 2000
rdf beta = 1.0
free_energy mode = standard
free_energy hs_functional = whitebeer
free_energy method = smf
free_energy integrated_strength_kernel = rdf
free_energy supplied_data = no
solution extrinsic_constraints: density: a = 0.15
solution extrinsic_constraints: density: b = 0.
```

Exact parsed dictionary

```
{
  "system": {
    "species": [
      "a",
      "b"
    ],
    "interactions": {
      "primary": {
        "aa": {
          "filename": "aa.txt"
        },
        "ab": {
          "filename": "ab.txt"
        },
        "bb": {
          "filename": "bb.txt"
        }
      }
    },
    "rdf": {
      "closure": {
        "aa": "hnc",
        "bb": "py",
        "ab": "py"
      },
      "alpha_max": 0.01,
      "r_max": 3.5,
      "max_iteration": 200000,
      "tolerance": 1e-06,
      "n_points": 2000,
    }
  }
}
```

```

    "beta": 1
  },
  "free_energy": {
    "mode": "standard",
    "hs_functional": "whitebeer",
    "method": "smf",
    "integrated_strength_kernel": "rdf",
    "supplied_data": "no"
  },
  "solution": {
    "extrinsic_constraints": {
      "density": {
        "a": 0.15,
        "b": 0
      }
    }
  }
}

```

Two-component EOS (24-12 case)

Active input lines

```

species = a, b
interactions primary: aa: filename = aa.txt
interactions primary: ab: filename = ab.txt
interactions primary: bb: filename = bb.txt
rdf closure aa = hnc
rdf closure bb = py
rdf closure ab = py
rdf alpha_max = 0.01
rdf r_max = 3.5
rdf max_iteration = 200000
rdf tolerance = 0.000001
rdf n_points = 2000
rdf beta = 1.0
free_energy mode = standard
free_energy hs_functional = whitebeer
free_energy method = smf
free_energy integrated_strength_kernel = rdf
free_energy supplied_data = no
solution extrinsic_constraints: density: a = 0.15
solution extrinsic_constraints: density: b = 0.

```

Exact parsed dictionary

```

{
  "system": {
    "species": [
      "a",
      "b"
    ],
    "interactions": {
      "primary": {
        "aa": {
          "filename": "aa.txt"
        },
        "ab": {
          "filename": "ab.txt"
        },
        "bb": {
          "filename": "bb.txt"
        }
      }
    },
    "rdf": {
      "closure": {
        "aa": "hnc",

```

```

    "bb": "py",
    "ab": "py"
  },
  "alpha_max": 0.01,
  "r_max": 3.5,
  "max_iteration": 200000,
  "tolerance": 1e-06,
  "n_points": 2000,
  "beta": 1
},
"free_energy": {
  "mode": "standard",
  "hs_functional": "whitebeer",
  "method": "smf",
  "integrated_strength_kernel": "rdf",
  "supplied_data": "no"
},
"solution": {
  "extrinsic_constraints": {
    "density": {
      "a": 0.15,
      "b": 0
    }
  }
}
}
}
}

```

Two-component binodal

Active input lines

```

species = a, b
interactions primary: aa: type = gs,  sigma = 1.0,  cutoff = 3.5,  epsilon = 2.0
interactions primary: ab: type = lj,  sigma = 1.0,  cutoff = 1.1224,  epsilon = 1.0
interactions primary: bb: type = mie, sigma = 1.0,  m = 24, n = 12,  cutoff = 2.5,  epsilon = 0.1
rdf closure aa = py
rdf closure ab = py
rdf closure bb = py
rdf alpha_max = 0.1
rdf max_iteration = 600000
rdf tolerance = 0.00001
rdf beta = 1.0
rdf r_max = 10.0
rdf n_points = 2000
free_energy ensemble = isochem
free_energy mode = standard
free_energy method = smf
free_energy hs_functional = whitebeer
free_energy integrated_strength_kernel = uniform
free_energy supplied_data = no
solution extrinsic_constraints: number_of_phases = 2
solution extrinsic_constraints: heterogeneous_pair = None
solution extrinsic_constraints: total_density_bound = 10
solution intrinsic_constraints: chemical_potential: a = 14

```

Exact parsed dictionary

```

{
  "system": {
    "species": [
      "a",
      "b"
    ],
    "interactions": {
      "primary": {
        "aa": {
          "type": "gs",
          "sigma": 1,

```

```

    "cutoff": 3.5,
    "epsilon": 2
  },
  "ab": {
    "type": "lj",
    "sigma": 1,
    "cutoff": 1.1224,
    "epsilon": 1
  },
  "bb": {
    "type": "mie",
    "sigma": 1,
    "m": 24,
    "n": 12,
    "cutoff": 2.5,
    "epsilon": 0.1
  }
},
"rdf": {
  "closure": {
    "aa": "py",
    "ab": "py",
    "bb": "py"
  },
  "alpha_max": 0.1,
  "max_iteration": 600000,
  "tolerance": 1e-05,
  "beta": 1,
  "r_max": 10,
  "n_points": 2000
},
"free_energy": {
  "ensemble": "isochem",
  "mode": "standard",
  "method": "smf",
  "hs_functional": "whitebeer",
  "integrated_strength_kernel": "uniform",
  "supplied_data": "no"
},
"solution": {
  "extrinsic_constraints": {
    "number_of_phases": 2,
    "heterogeneous_pair": "None",
    "total_density_bound": 10
  },
  "intrinsic_constraints": {
    "chemical_potential": {
      "a": 14
    }
  }
}
}
}

```

Two-component structural analysis

Active input lines

```

species = a, b
interactions primary: aa: filename = aa.txt
interactions primary: ab: filename = ab.txt
interactions primary: bb: filename = bb.txt
rdf closure aa = hnc
rdf closure bb = py
rdf closure ab = py
rdf alpha_max = 0.1

```

```

rdf max_iteration = 400000
rdf tolerance = 0.00001
rdf beta = 1.0
rdf r_max = 10.0
rdf n_points = 2000

```

Exact parsed dictionary

```

{
  "system": {
    "species": [
      "a",
      "b"
    ],
    "interactions": {
      "primary": {
        "aa": {
          "filename": "aa.txt"
        },
        "ab": {
          "filename": "ab.txt"
        },
        "bb": {
          "filename": "bb.txt"
        }
      }
    },
    "rdf": {
      "closure": {
        "aa": "hnc",
        "bb": "py",
        "ab": "py"
      },
      "alpha_max": 0.1,
      "max_iteration": 400000,
      "tolerance": 1e-05,
      "beta": 1,
      "r_max": 10,
      "n_points": 2000
    }
  }
}

```

PART V — Source Reference, Outputs, and Package Maintenance

23. Module-by-Module Reference

The following catalogue is generated from the Python source tree. “Demonstrated” means the module appears directly in the bundled Example_workflow scripts or in their immediate data path; other modules are included for developer orientation.

[cdft_solver/calculators/c_2_analysis/c_analysis.py](#)

Structural/reference c(r), sigma, B2, G(r), Delta-c analysis.

Callable	Source description / role
<code>solve_oz_kspace(h_k, densities, eps)</code>	Solve multi-component OZ equation in k-space:
<code>solve_oz_matrix(c_r, r, densities)</code>	

multi_component_oz_solver_alpha(r, pair_closures, densities, u_matrix, sigma_matrix, c_update_flag, c_initial, n_iter, tol, alpha_rdf_max, gamma_initial)	Multi-component OZ solver with adaptive alpha-mixing
wca_split(r, u)	Returns repulsive part of u(r) using WCA splitting.
detect_sigma_from_gr(r, g, g_tol, min_width)	Detect hard-core diameter ONLY if g(r) is strictly zero
boltzmann_potential_from_gr(g, beta, g_min)	$u(r) = -\ln g(r)$ with numerical protection
detect_first_minimum_near_core(r, u_ij, sigma)	Detect the first local minimum of u(r) after the hard core.
find_key_recursive(d, key)	
plot_u_matrix(r, u_matrix, species, outdir, filename)	Plot and export $u_{ij}(r)$ for all species pairs.
c_analysis(ctx, rdf_config, supplied_data, densities, filename_prefix, export_plot, export_json)	Multistate Isotropic Boltzmann / IBI inversion

[cdft_solver/calculators/coexistence_densities/calculator_coexistence_densities_isochem.py](#)

Multiphase isochemical coexistence solver.

Callable	Source description / role
deep_get(data, key, default)	Recursively search for key in nested dicts/lists.
deep_get_all(data, key)	Return all matches for key in nested structure.
build_thermodynamics_from_fe_res(fe_res)	Given a free-energy result dictionary (from JSON),
coexistence_densities_isochem(ctx, config_dict, fe_res, supplied_data, max_outer_iters, tol_outer, tol_solver, verbose)	Iso-chemical coexistence solver using ONLY a supplied dictionary.

[cdft_solver/calculators/coexistence_densities/calculator_coexistence_densities_isocore.py](#)

Canonical/isocore coexistence solver.

Callable	Source description / role
deep_get(data, key, default)	Recursively search for key in nested dicts/lists.
deep_get_all(data, key)	Return all matches for key in nested structure.
build_thermodynamics_from_fe_res(fe_res)	Given a free-energy result dictionary (from JSON),
coexistence_densities_isocore(ctx, config_dict, fe_res, supplied_data, max_outer_iters, tol_outer, tol_solver, verbose)	Iso-chemical coexistence solver using ONLY a supplied dictionary.

[cdft_solver/calculators/coexistence_densities/kernel_generator.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
build_strength_kernel(ctx, config, supplied_data, densities, kernel_type)	Dispatcher for integrated strength kernel.

[cdft_solver/calculators/dispersion_relation/dispersion_relation.py](#)

Dispersion-relation calculation from structural input.

Callable	Source description / role
<code>dispersion_relation(ctx, rdf_config, densities, diffusivity, supplied_data, export, plot, filename_prefix)</code>	Compute dispersion relation $\omega(k)$ using RDF input.

[cdft_solver/calculators/free_energy_hard_core/hard_core_bulk_rosenfeld.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>hard_core_bulk_rosenfeld(ctx, hc_data, export_json, filename)</code>	Computes the hard-core contribution to the free energy for a multi-species system.

[cdft_solver/calculators/free_energy_hard_core/hard_core_bulk_whitebeer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>hard_core_bulk_whitebeer(ctx, hc_data, export_json, filename)</code>	White Bear FMT (true form) for colloids + Free Volume Theory for polymers.

[cdft_solver/calculators/free_energy_hard_core/hard_core_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>hard_core_planer(ctx, hc_data, export_json, filename)</code>	Computes the hard-core free energy (FMT-like) for multi-species system

[cdft_solver/calculators/free_energy_hybrid/hybrid.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>hybrid(ctx, hc_data, export_json, filename)</code>	Computes hybrid free energy:

[cdft_solver/calculators/free_energy_hybrid/hybrid_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>hybrid_planer(ctx)</code>	Computes the hard-core contribution to the free energy for a multi-species system.

[cdft_solver/calculators/free_energy_hybrid_lattice/free_energy_lattice.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>free_energy_lattice(ctx, hc_data, export_json, filename)</code>	Computes lattice free energy:

[cdft_solver/calculators/free_energy_ideal/ideal.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>ideal(ctx, hc_data, export_json, filename)</code>	Computes the symbolic ideal (entropic) part of the free energy for a multi-species system.

[cdft_solver/calculators/free_energy_mean_field/EMF.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>free_energy_EMF(ctx, hc_data, export_json, filename)</code>	

[cdft_solver/calculators/free_energy_mean_field/EMF_z.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>free_energy_EMF_z(ctx, hc_data, export_json, filename)</code>	Enhanced Mean-Field (EMF) two-point free-energy kernel:

[cdft_solver/calculators/free_energy_mean_field/SMF.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>free_energy_SMF(ctx, hc_data, export_json, filename)</code>	Computes the symbolic mean-field (SMF) free energy for a multi-species system

[cdft_solver/calculators/free_energy_mean_field/SMF_z.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>free_energy_SMF_z(ctx, hc_data, export_json, filename)</code>	Standard Mean-Field (SMF) two-point free-energy kernel:

[cdft_solver/calculators/free_energy_mean_field/mean_field.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
mean_field(ctx, hc_data, system_config, export_json, filename)	Unified wrapper for symbolic free energy models (EMF, SMF, VOID).

[cdft_solver/calculators/free_energy_mean_field/mean_field_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
mean_field_planer(ctx, hc_data, system_config, export_json, filename)	Unified wrapper for symbolic free energy models (EMF, SMF, VOID).

[cdft_solver/calculators/free_energy_mean_field/void.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
free_energy_void(ctx, hc_data, export_json, filename)	Computes the symbolic cavity mean-field (CMF) free energy for a multi-species system

[cdft_solver/calculators/free_energy_mean_field/void_z.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
free_energy_void_z(ctx, hc_data, export_json, filename)	Cavity Mean-Field (CMF) two-point free-energy kernel:

[cdft_solver/calculators/g_alpha_r/delta_c.py](#)

Direct-correlation-difference kernel route.

Callable	Source description / role
solve_oz_kspace(h_k, densities, eps)	Solve multi-component OZ equation in k-space:
plot_u_matrix(r, u_matrix, species, outdir, filename)	Plot and export $u_{ij}(r)$ for all species pairs.
process_supplied_rdf(supplied_data, species, r_grid)	Uses canonical pair keys like 'AA', 'AB', ...
hankel_forward_dst(f_r, r)	
hankel_inverse_dst(k, Fk, r)	
hankel_transform_matrix_fast(f_r_matrix, r)	
inverse_hankel_transform_matrix_fast(f_k_matrix, k, r)	
solve_oz_matrix(c_r, r, densities)	

multi_component_oz_solver_alpha(r, pair_closures, densities, u_matrix, sigma_matrix, c_update_flag, c_initial, n_iter, tol, alpha_rdf_max, gamma_initial)	Multi-component OZ solver with adaptive alpha-mixing
apply_adaptive_ylim(ax, ydata, limit, clip)	
plot_matrix_quantity(r, quantity, u_matrix, species, title_prefix, filename, plots_dir)	
wca_split(r, u)	Returns repulsive part of u(r) using WCA splitting.
detect_sigma_from_gr(r, g, g_tol, min_width)	Detect hard-core diameter ONLY if g(r) is strictly zero
boltzmann_potential_from_gr(g, beta, g_min)	$u(r) = -\ln g(r)$ with numerical protection
detect_first_minimum_near_core(r, u_ij, sigma)	
find_key_recursive(d, key)	
delta_c_alpha(ctx, rdf_config, densities, supplied_data, export, plot, filename_prefix)	Fully dictionary-driven isotropic RDF solver

[cdft_solver/calculators/g_alpha_r/rdf_alpha_r.py](#)

Coupling-parameter RDF/interaction kernel construction.

Callable	Source description / role
solve_oz_kspace(h_k, densities, eps)	Solve multi-component OZ equation in k-space:
plot_u_matrix(r, u_matrix, species, outdir, filename)	Plot and export $u_{ij}(r)$ for all species pairs.
process_supplied_rdf(supplied_data, species, r_grid)	Uses canonical pair keys like 'AA', 'AB', ...
hankel_forward_dst(f_r, r)	
hankel_inverse_dst(k, Fk, r)	
hankel_transform_matrix_fast(f_r_matrix, r)	
inverse_hankel_transform_matrix_fast(f_k_matrix, k, r)	
solve_oz_matrix(c_r, r, densities)	
multi_component_oz_solver_alpha(r, pair_closures, densities, u_matrix, sigma_matrix, c_update_flag, c_initial, n_iter, tol, alpha_rdf_max, gamma_initial)	Multi-component OZ solver with adaptive alpha-mixing
apply_adaptive_ylim(ax, ydata, limit, clip)	
plot_matrix_quantity(r, quantity, u_matrix, species, title_prefix, filename, plots_dir)	
wca_split(r, u)	Returns repulsive part of u(r) using WCA splitting.
detect_sigma_from_gr(r, g, g_tol, min_width)	Detect hard-core diameter ONLY if g(r) is strictly zero
boltzmann_potential_from_gr(g, beta, g_min)	$u(r) = -\ln g(r)$ with numerical protection
detect_first_minimum_near_core(r, u_ij, sigma)	Detect the first local minimum of u(r) after the hard core.
find_key_recursive(d, key)	

rdf_alpha_r(ctx, rdf_config, densities, supplied_data, export, plot, filename_prefix)	Fully dictionary-driven isotropic RDF solver
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[cdft_solver/calculators/integrated_strength/integrated_strength_planer_kernel.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
vij_planer_kernel(ctx, config, kernel_data, u_data, export_json, filename, plot)	Compute planar MF coupling:

[cdft_solver/calculators/integrated_strength/integrated_strength_radial_kernel.py](#)

Pairwise integrated strength v_{ij} from radial kernels.

Callable	Source description / role
vij_radial_kernel(ctx, config, kernel, supplied_data, export_json, filename)	Compute:

[cdft_solver/calculators/one_d_profile_iterator/kernel_generator_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
build_strength_kernel_planer(ctx, config, densities, supplied_data, kernel_type)	Build planar strength kernel.

[cdft_solver/calculators/one_d_profile_iterator/one_d_profile_iterator_box.py](#)

One-dimensional density-profile iteration.

Callable	Source description / role
deep_get(data, key, default)	Recursively search for key in nested dicts/lists.
deep_get_all(data, key)	Return all matches for key in nested structure.
build_thermodynamics_from_fe_res(fe_res)	Given a free-energy result dictionary (from JSON),
one_d_profile_iterator_box(ctx, config, export_json, export_plots, filename)	Required solvers (install via pip if needed).

[cdft_solver/calculators/phase_finder/thermodynamic_potentials_isochem.py](#)

Grand/isochemical thermodynamic state evaluation.

Callable	Source description / role
deep_get(data, key, default)	Recursively search for key in nested dicts/lists.
deep_get_all(data, key)	Return all matches for key in nested structure.

build_thermodynamics_from_fe_res(fe_res)	Given a free-energy result dictionary (from JSON),
find_key_recursive(d, key)	
evaluate_grand_canonical_state(ctx, config_dict, fe_res, supplied_data, export, output_file, verbose)	Grand canonical evaluation at given fixed densities.

[cdft_solver/calculators/phase_finder/thermodynamic_potentials_isocore.py](#)

Canonical state evaluation from densities.

Callable	Source description / role
deep_get(data, key, default)	Recursively search for key in nested dicts/lists.
deep_get_all(data, key)	Return all matches for key in nested structure.
find_key_recursive(d, key)	
build_thermodynamics_from_fe_res(fe_res)	Given a free-energy result dictionary (from JSON),
evaluate_canonical_state(ctx, config_dict, fe_res, supplied_data, export, output_file, verbose)	Canonical evaluation:

[cdft_solver/calculators/radial_distribution_function/builtin.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
py_closure(r, gamma, u, sigma_ab)	Percus-Yevick (PY) closure.
hnc_closure(r, gamma, u, sigma_ab)	
hybrid_closure(r, gamma, u, sigma_ab)	

[cdft_solver/calculators/radial_distribution_function/closure.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
closure_update_c_matrix(gamma_r_matrix, r, pair_closures, u_matrix, sigma_matrix, closure_registry)	pair_closures[a,b] can be:

[cdft_solver/calculators/radial_distribution_function/rdf_planer.py](#)

Planar correlation solver.

Callable	Source description / role
plot_correlation_functions(gamma_r, c_r, r_grid, z_grid, Ns, Nz, Nr, ctx, plot)	Plot and save RDF results.
hankel_transform_2d(f_r, r, k_grid, J0)	Apply Hankel transform for cylindrical coordinates along r.

inverse_hankel_transform_2d(F_k, r, k_grid, J0)	
solve_oz_matrix_2d(c_r, RHO, r_grid, k_grid, J0, Ns, Nz, Nr)	
solve_oz_matrix_2d(c_r_matrix, RHO, r, k_grid, J0, Ns, Nz, Nr)	2D OZ solver along r for each plane z_i, z_j using the C solver.
rdf_planer(ctx, rdf_config, densities, supplied_data, export, plot, filename_prefix)	

[cdft_solver/calculators/radial_distribution_function/rdf_radial.py](#)

3D isotropic multicomponent OZ/RDF solver.

Callable	Source description / role
solve_oz_kspace(h_k, densities, eps)	Solve multi-component OZ equation in k-space:
plot_u_matrix(r, u_matrix, species, outdir, filename)	Plot and export u_ij(r) for all species pairs.
process_supplied_rdf(supplied_data, species, r_grid)	Uses canonical pair keys like 'AA', 'AB', ...
hankel_forward_dst(f_r, r)	
hankel_inverse_dst(k, Fk, r)	
hankel_transform_matrix_fast(f_r_matrix, r)	
inverse_hankel_transform_matrix_fast(f_k_matrix, k, r)	
solve_oz_matrix(c_r, r, densities)	
multi_component_oz_solver_alpha(r, pair_closures, densities, u_matrix, sigma_matrix, c_update_flag, c_initial, n_iter, tol, alpha_rdf_max)	Multi-component OZ solver with adaptive alpha-mixing
apply_adaptive_ylim(ax, ydata, limit, clip)	
plot_matrix_quantity(r, quantity, u_matrix, species, title_prefix, filename, plots_dir)	
find_key_recursive(d, key)	
rdf_radial(ctx, rdf_config, densities, supplied_data, export, plot, filename_prefix)	Fully dictionary-driven isotropic RDF solver

[cdft_solver/calculators/radial_distribution_function/rdf_two_d.py](#)

2D isotropic OZ/RDF solver.

Callable	Source description / role
solve_oz_kspace(h_k, densities, eps)	Solve multi-component OZ equation in k-space:
plot_u_matrix(r, u_matrix, species, outdir, filename)	Plot and export u_ij(r) for all species pairs.
process_supplied_rdf(supplied_data, species, r_grid)	Uses canonical pair keys like 'AA', 'AB', ...
build_k(r)	
build_bessel_matrix(k, r)	

<code>hankel_forward_core(f_r, r, J0, Nk, Nr)</code>	
<code>hankel_forward_2d(f_r, r, k, J0)</code>	
<code>hankel_inverse_2d(Fk, k, r, J0)</code>	
<code>hankel_transform_matrix(f_r_matrix, r, k, J0)</code>	
<code>inverse_hankel_transform_matrix(f_k_matrix, r, k, J0)</code>	
<code>solve_oz_kspace_numba(c_k, densities)</code>	
<code>multi_component_oz_solver_alpha(r, pair_closures, densities, u_matrix, sigma_matrix, n_iter, tol, alpha_rdf_max)</code>	
<code>apply_adaptive_ylim(ax, ydata, limit, clip)</code>	
<code>plot_matrix_quantity(r, quantity, u_matrix, species, title_prefix, filename, plots_dir)</code>	
<code>find_key_recursive(d, key)</code>	
<code>rdf_2d(ctx, rdf_config, densities, supplied_data, export, plot, filename_prefix)</code>	Fully dictionary-driven isotropic RDF solver

[cdft_solver/calculators/radial_distribution_function/utils.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>safe_exp(x, xmin, xmax)</code>	Numerically safe exponential.

[cdft_solver/calculators/rdf_inversion/data_analyser_examples/plotter_integrated_strength.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>find_common_crossing(curves, r, r_guess)</code>	

[cdft_solver/calculators/rdf_inversion/data_analyser_examples/plotter_structure_factor.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>structure_factor_ij(k, r, g_ij, rho_i, rho_j)</code>	Partial structure factor $S_{ij}(k)$

[cdft_solver/calculators/rdf_inversion/rdf_inversion_advance.py](#)

Multistate IBI/Boltzmann inversion plus reference diagnostics.

Callable	Source description / role
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<code>solve_oz_kspace(h_k, densities, eps)</code>	Solve multi-component OZ equation in k-space:
<code>solve_oz_matrix(c_r, r, densities)</code>	
<code>multi_component_oz_solver_alpha(r, pair_closures, densities, u_matrix, sigma_matrix, c_update_flag, c_initial, n_iter, tol, alpha_rdf_max, gamma_initial)</code>	Multi-component OZ solver with adaptive alpha-mixing
<code>wca_split(r, u)</code>	Returns repulsive part of $u(r)$ using WCA splitting.
<code>detect_sigma_from_gr(r, g, g_tol, min_width)</code>	Detect hard-core diameter ONLY if $g(r)$ is strictly zero
<code>boltzmann_potential_from_gr(g, beta, g_min)</code>	$u(r) = -\ln g(r)$ with numerical protection
<code>detect_first_minimum_near_core(r, u_ij, sigma)</code>	Detect the first local minimum of $u(r)$ after the hard core.
<code>clean_gr_tail(r, g, prominence, alpha, noise_tol, min_persistence)</code>	Physically motivated RDF tail cleaning.
<code>process_supplied_rdf_multistate(supplied_data, species, r_grid)</code>	
<code>find_key_recursive(d, key)</code>	
<code>plot_u_matrix(r, u_matrix, species, outdir, filename)</code>	Plot and export $u_{ij}(r)$ for all species pairs.
<code>boltzmann_inversion_advance(ctx, rdf_config, supplied_data, filename_prefix, export_plot, export_json)</code>	Multistate Isotropic Boltzmann / IBI inversion

[cdft_solver/calculators/total_free_energy/free_energy_exporter.py](#)

Serialize symbolic total free energy.

Callable	Source description / role
<code>_serialize_symbol(sym)</code>	Serialize a SymPy symbol safely.
<code>_serialize_expression(expr)</code>	Serialize a SymPy expression (lossless).
<code>free_energy_exporter(ctx, total_fe, filename, system_config, indent)</code>	Export merged total free energy to a JSON file.

[cdft_solver/calculators/total_free_energy/total_free_energy.py](#)

Free-energy dispatcher and symbolic component merger.

Callable	Source description / role
<code>unify_symbols(components)</code>	Unify SymPy symbols across multiple free-energy components
<code>extract_sigma_matrix(hc_data)</code>	
<code>sigma_is_zero(sigma_matrix, tol)</code>	
<code>merge_free_energies(components)</code>	Merge multiple free-energy components into one symbolic object,
<code>find_key_recursive(d, key)</code>	
<code>total_free_energy(ctx, hc_data, system_config, export_json, filenames)</code>	Unified free-energy dispatcher.

[cdft_solver/calculators/virial_analysis/second_virial.py](#)

Second-virial analysis.

Callable	Source description / role
find_key_recursive(d, key)	
second_virial(ctx, virial_config, on, export, filename_prefix, r_max_factor, nr, n_lambda, beta_scale)	Research-grade computation of:

[cdft_solver/calculators/virial_analysis/second_virial_scale_calibration.py](#)

Calibrate potential scales against target B2.

Callable	Source description / role
find_zero_crossing(r, u)	
compute_sigma_BH(scale, r, u, tol, hc_threshold)	Compute Barker–Henderson diameter with robust hard-core detection.
find_key_recursive(d, key)	
compute_B2_scaled(scale, r, u_total)	Second virial coefficient using a globally scaled total potential:
unpack_scale_vector(scale_vec, pair_list, n)	
second_virial_scale_calibration(ctx, virial_config, B2_target, on, export, filename_prefix, r_max_factor, nr, beta_scale)	Calibrate pairwise scalar factors f_{ij} such that

[cdft_solver/engines/bulk.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
bulk_executor(ctx)	Bulk executor for Density Functional Theory simulations.

[cdft_solver/engines/one_d_profile.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
one_d_profile_executor(ctx)	Bulk executor for Density Functional Theory simulations.

[cdft_solver/engines/rdf_bulk.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
rdf_bulk_executor(ctx)	Bulk executor for Density Functional Theory simulations.

[cdft_solver/executor_dft_main.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>parse_input_keywords(input_text)</code>	Parse ensemble, method, task, and profile from an input file.
<code>main()</code>	

[cdft_solver/generators/density_weights/fmt_weights_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>fmt_weights_planer(ctx, data_dict, grid_properties, export_json, filename, plot)</code>	Generate 1D FMT weight functions for cylindrical geometry.

[cdft_solver/generators/density_weights/mf_weights_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>mf_weights_planer(ctx, data_dict, grid_properties, export_json, filename, plot)</code>	Dictionary-driven MF cylindrical kernel generator.

[cdft_solver/generators/grids_properties/bulk_rho_mue_planer.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>bulk_rho_mue_planer(ctx, thermodynamic_parameter, r_space_coordinates, export_json, filename, plot)</code>	Assign bulk density and chemical potential values to an r-space grid

[cdft_solver/generators/grids_properties/external_potential_grid.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>external_potential_grid(ctx, data_dict, grid_properties, export_json, filename, plot)</code>	Compute external potentials from a dictionary and a spatial grid.

[cdft_solver/generators/grids_properties/k_and_r_space_box.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>r_k_space_box(ctx, data_dict, export_json, filename)</code>	Generate r-space and k-space grids from a dictionary.

[cdft_solver/generators/grids_properties/k_and_r_space_cylindrical.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>r_k_space_cylindrical(ctx, data_dict, export_json, filename)</code>	Generate cylindrical r-space and k-space grids from a dictionary.

[cdft_solver/generators/grids_properties/k_and_r_space_spherical.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>r_k_space_spherical(ctx, data_dict, export_json, filename)</code>	Generate spherical r-space and k-space grids from a dictionary.

[cdft_solver/generators/parameters/advance_dictionary.py](#)

Hierarchical .in parser and JSON exporter.

Callable	Source description / role
<code>super_dictionary_creator(ctx, input_file, base_dict, export_json, filename, super_key_name)</code>	Universal dictionary builder from hierarchical input.

[cdft_solver/generators/potential/pair_potential_isotropic.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
<code>pair_potential_isotropic(specific_pair_potential)</code>	

[cdft_solver/generators/potential/pair_potential_isotropic_default.py](#)

Built-in analytic isotropic potential implementations.

Callable	Source description / role
<code>zero_potential(p)</code>	
<code>hard_core(p)</code>	
<code>lj(p)</code>	
<code>gaussian(p)</code>	
<code>generalized_gaussian(p)</code>	Generalized Gaussian / GEM-n potential
<code>mie(p)</code>	

custom_1(p)	
custom_3(p)	
wca(p)	
salj(p)	
nvrn(p)	
ma(p)	

[cdft_solver/generators/potential/pair_potential_isotropic_registry.py](#)

Potential registry and factory lookup.

Callable	Source description / role
pair_potential_isotropic(specific_pair_potential)	
register_isotropic_pair_potential(potential_type, factory_fn, overwrite)	Register an isotropic pair potential factory.
get_isotropic_pair_potential_factory(potential)	Resolve factory from registry.

[cdft_solver/generators/potential_splitter/hc.py](#)

Hard-core detection, effective sigma matrix, HC potential generation.

Callable	Source description / role
hard_core_potentials(ctx, input_data, grid_points, file_name_prefix, export_files)	Hard-core detection module using a dictionary input instead of a JSON file.

[cdft_solver/generators/potential_splitter/mf.py](#)

Mean-field interaction conversion and potential generation.

Callable	Source description / role
meanfield_potentials(ctx, input_data, grid_points, file_name_prefix, export_files)	Mean-field potential generator using a dictionary input.

[cdft_solver/generators/potential_splitter/mf_registry.py](#)

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
register_potential_converter(potential_type, converter_fn, overwrite)	Register a potential conversion function.
convert_potential_via_registry(potential)	Convert a potential dictionary using the registry.
_hc_to_zero(pot)	

_lj_to_salj(pot)	
_mie_to_ma(pot)	

cdft_solver/generators/potential_splitter/raw.py

Unsplitted sum of primary/secondary/tertiary pair interactions.

Callable	Source description / role
raw_potentials(ctx, input_data, grid_points, file_name_prefix, export_files)	

cdft_solver/generators/potential_splitter/total.py

Combine hard-core and mean-field potential representations.

Callable	Source description / role
load_json_or_dict(obj)	Accepts either:
reconstruct_potential(pot_dict, u_key)	Convert JSON potential entry to NumPy arrays.
find_key_recursive(obj, key)	Recursively find ALL occurrences of 'key' in nested dict/list structures.
build_hc_flag_map(species, flag_matrix)	Build map: pair -> 0/1
interpolate_to_grid(r_src, u_src, r_target)	Interpolate u_src(r_src) onto r_target.
total_potentials(ctx, hc_source, mf_source, file_name_prefix, export_files)	Assemble hard-core + mean-field potentials with:

cdft_solver/generators/supplied_data/process_supplied_data.py

Materializes filename-backed numerical supplied data.

Callable	Source description / role
find_key_recursive(d, key)	
_load_xy_file(path)	
materialize_supplied_data(obj, base_dir)	Recursively walk supplied_data and replace
_to_serializable(obj)	
_plot_materialized(obj, plots_dir, prefix)	
process_supplied_data(ctx, config, export_json, export_plot)	

cdft_solver/generators/supplied_data/process_supplied_potential.py

Registers/replaces supplied potential data.

Callable	Source description / role
----------	---------------------------

supplied_potential_factory(r_data, U_data)	Return callable interpolated potential V(r).
wca_split(r, U)	Split a potential into repulsive (soft) and attractive parts (WCA).
has_hard_core(U, threshold)	Heuristic to detect a hard-core in a potential.
find_key_recursive(d, key)	Recursively find the first occurrence of a key in a nested dict.
replace_primary_potential_recursive(config, pair, new_name)	Recursively search for a 'primary' section inside 'potentials' and
register_supplied_potentials(ctx, config, supplied_data, export_json, export_plot)	Register supplied potentials as callable functions, split if needed,

cdft_solver/utils.py

Supporting module; inspect function signature and source before using in a new public workflow.

Callable	Source description / role
create_unique_scratch_dir(base_name)	Create a unique scratch directory in the current working directory.
class ExecutionContext	Class defined in this module.

24. Output File and Array Shape Reference

Workflow	Primary return	Primary exported files	Key shapes
Potential preprocessing	hc_data / mf_data / raw_data / total_data	hc.json, mf.json, raw.json, total.json	pair r,U vectors; sigma/flag N×N
Free energy	dict with SymPy expression/function	fe_id/mf/hc/hybrid + fe_total	variables length N+N ² (for MF symbols)
Canonical state	state dict	state_from_density.json optional	densities N; vij N×N; mu N
RDF radial	pair structural output	prefix JSON/plots depending function	g,c,gamma N×N×Nr
Inversion	(u_matrix, sigma_matrix)	prefix_potential.json + plots/diagnostics	u N×N×Nr; sigma N×N
c_analysis	(u_matrix, sigma_matrix)	result_sigma_analysis, delta_B2, result_G_of_r, delta_c, prefix_potential	many N×N×Nr arrays
Coexistence	dict	workflow-controlled text/JSON/plot	rhos nphases×N; vij nphases×N×N

24.1 Pair indexing convention

```
species = [a, b]
index 0 -> a
index 1 -> b

matrix[0,0] -> aa
matrix[0,1] -> ab
```

```
matrix[1,0] -> ba (normally symmetric with ab)
matrix[1,1] -> bb
```

24.2 Radial grid convention in RDF/inversion modules

For r_{max} and n_{points} , the standard 3D radial grid is constructed with $dr = r_{\text{max}}/(n_{\text{points}}+1)$ and $r = dr * [1,2,...,n_{\text{points}}]$. Therefore $r=0$ is not an explicit structural grid point, although some integration helpers insert a zero anchor internally when calculating effective diameters.

25. Quick Reference: Which Workflow Do I Use?

Scientific question	Primary workflow / function	Minimum input blocks
What is $P(\rho)$ for one component?	preprocessing + evaluate_canonical_state density loop	species, interaction, rdf, free_energy, solution density
What is $P(\rho_t)$ at fixed binary composition?	binary canonical state loop	species, all pair interactions, closures, free_energy, both densities
What potential reproduces a supplied RDF?	process_supplied_data + boltzmann_inversion_advance	species, rdf settings, supplied_data states
What are $g(r)$, $c(r)$, reference sigma, Delta B2 and Delta c?	c_analysis	species, interactions, closures, density array
What is v_{ij} by uniform/B2/RDF/delta-c route?	build_strength_kernel + vij_radial_kernel, often called indirectly	interactions, rdf, free_energy integrated_strength_kernel
What are coexisting densities?	preprocessing + coexistence_densities_isochem	species, interactions, free_energy, phase and intrinsic constraints
What is one state's μ and pressure?	evaluate_canonical_state	full free energy plus density for every species

26. Recommended Improvements Before Public Release

These recommendations follow directly from the source audit and are included because they materially improve the reliability of the manual/data contract for other users.

32. Synchronize package version between setup.py and pyproject.toml.
33. Add sympy, numba and tqdm to declared dependencies if these workflows are public API; document pyfftw only for any retained legacy engine path.
34. Add work_dir to ExecutionContext and resolve all filename-backed data relative to the input file directory rather than Path.cwd().
35. Define/import SuppliedDataError in process_supplied_data.
36. Standardize tolerance spelling and keep a backward-compatible alias for tolerance if old inputs must continue to run.
37. Standardize free-energy comments/documentation to the actual accepted keys: mode standard/hybrid/hybrid_lattice and method smf/emf/void.
38. Normalize species to a list immediately after parsing and define an explicit pair-key convention that supports multi-character species names.
39. Add schema validation after super_dictionary_creator so missing pair closures/densities and invalid values are caught before expensive solvers start.
40. Write all workflow-created output files into ctx.scratch_dir rather than occasionally into the current working directory.

41. Add small regression tests that parse every example input, run a low-resolution potential/free-energy setup, and check output keys/shapes.
42. Replace or repair the legacy executor/engine imports, or clearly mark them deprecated so users start from the direct API documented here.

27. Closing Summary

The supplied cDFT_solver snapshot already contains the pieces for a broad cDFT/integral-equation workflow: hierarchical input parsing, analytic/tabulated potentials, hard-core and mean-field decomposition, OZ closures, effective-potential inversion, symbolic free energies, integrated-strength routes, canonical thermodynamics, and coexistence. The most important practical requirement is to preserve the data contract between these pieces. For that reason, this manual treats every input line, parsed dictionary, numerical array, and exported JSON object as part of the scientific method rather than as incidental software plumbing.

For new calculations, the recommended sequence is: validate the .in file by exporting input_system.json; inspect potential decomposition and sigma/flag matrices; build the free energy once; then run a clearly scoped state, structural, inversion, or coexistence workflow while saving the relevant kernel/interaction quantities alongside the final observable.

PART VI — Advanced Extensibility, Supplied Data, and Multistate Inversion

This extended part supplements the preceding source-audited manual. It focuses on the package features that are easiest to miss when reading only the bundled workflows: user-defined closures, runtime-registered potentials, file-backed potentials, mixed supplied/computed structural data, and effective-potential inversion constrained by several thermodynamic state points.

Source basis — The descriptions below are based on the active modules `closure.py`, `registry.py`, `builtin.py`, `rdf_radial.py`, `pair_potential_isotropic_registry.py`, `pair_potential_isotropic.py`, `pair_potential_isotropic_default.py`, `raw.py`, `hc.py`, `mf.py`, `mf_registry.py`, `process_supplied_data.py`, `process_supplied_potential.py`, `rdf_inversion_advance.py`, `c_analysis.py`, `external_potential_grid.py`, and the bundled `Example_workflow` inputs/scripts.

28. Advanced Extension and Data-Injection Overview

The package supports several independent mechanisms for introducing data or physics from outside the built-in definitions. The correct mechanism depends on what is being supplied and at what stage it should enter the computation.

Mechanism	Where it enters	Internal object	Best use
Built-in analytical potential	interactions ... type = ...	potential factory -> V(r)	standard LJ/Mie/Gaussian/HC/GEM-style interactions
Tabulated pair potential	interactions ... filename = ...	interpolated r,U table	simulation-derived, inverted, fitted, or externally generated pair potential
Runtime custom potential	Python potential registry	user factory: dict -> callable V(r)	new functional form without editing package source
Manual closure	rdf.closure after parsing or closure registry	string name or Python callable	test a new OZ closure on selected pairs
Generic supplied file data	supplied_data + process_supplied_data	nested x/y arrays	RDF state data and other numerical series
Constrained structural data	rdf_radial supplied_data	pair r/g arrays + fixed mask	hold selected RDFs fixed while solving the rest
Multistate inversion data	boltzmann_inversion_advance	states -> densities, beta, g_target, fixed_mask	infer a shared effective potential from one or many states

Important terminology — “Supplied data” is not a single universal schema in this source snapshot. The generic loader produces x/y; advanced inversion accepts x/y or r/g; `rdf_radial` constrained projection expects r/g; supplied-potential registration expects r/U. The later compatibility table makes these differences explicit.

29. Adding a Manual Closure Through the Workflow

The closure dispatcher was written to accept either a built-in closure name or a Python callable on a pair-by-pair basis. This is the cleanest extension point for testing a new liquid-state closure without modifying the package source files.

29.1 Built-in closure path

The text input normally creates a nested closure dictionary such as the following:

```
species = a, b
rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = hybrid
```

After `super_dictionary_creator`, this becomes `system["system"]["rdf"]["closure"] = {"aa":"hnc", "ab":"py", "bb":"hybrid"}`. `rdf_radial`, `c_analysis`, and `boltzmann_inversion_advance` convert that dictionary into an $N \times N$ object matrix and mirror each $i \leq j$ selection to j, i .

29.2 Exact callable signature required by `closure_update_c_matrix`

Required custom-closure interface

```
def user_closure(*, r, gamma, u, sigma_ab=None):
    # r      : ndarray, shape (Nr,)
    # gamma  : ndarray, shape (Nr,)
    # u      : ndarray, shape (Nr,)
    # sigma_ab : float or None
    # Return c(r) on the same grid and with the same shape.
    c_r = ...
    return c_r
```

Argument	Meaning	Critical detail
r	radial grid	same 1D grid used by the OZ iteration
gamma	current indirect correlation $\gamma=h-c$	one value per radial point
u	pair potential passed to the closure	in active radial workflows it is already scaled as $\beta \cdot u$ for the current state
sigma_ab	effective hard-core/reference diameter	can be None or zero; do not assume a hard core
return	new direct correlation $c(r)$	must be a numerical 1D array matching <code>gamma.shape</code>

Keyword calling convention — The dispatcher invokes a callable as `closure(r=r, gamma=gamma, u=u, sigma_ab=...)`. A function with incompatible keyword names will fail even if the numerical formula itself is valid.

29.3 Recommended method: replace one closure after parsing

Because a Python function cannot be represented in the plain-text `.in` file, the most transparent research workflow is: keep a readable placeholder closure in the input file, parse the file, then replace the desired pair entry with the callable before the structural calculation begins.

Manual closure injected into the ordinary workflow

```
import numpy as np
from cdft_solver.generators.parameters.advance_dictionary import super_dictionary_creator
```

```
# HNC-equivalent formula used here only to demonstrate the software interface.
def my_hnc(*, r, gamma, u, sigma_ab=None):
    return np.exp(-u + gamma) - gamma - 1.0

system = super_dictionary_creator(
    ctx=ctx,
    export_json=True,
    filename='input_system_before_python_override.json',
)

# .in file may say: rdf closure ab = py
system['system']['rdf']['closure']['ab'] = my_hnc

# The normal workflow can now be called.
# rdf_radial(...), c_analysis(...), or boltzmann_inversion_advance(...)
```

The callable survives because the solver stores pair_closures in a NumPy object array. closure_update_c_matrix checks callable(closure) before trying a string registry lookup.

29.4 Alternative method: register a new closure name globally

The built-in registry is the dictionary CLOSURE_REGISTRY. It contains PY, HNC, and HYBRID. A user can add another entry before the solver builds its pair-closure matrix.

Register and select a named custom closure

```
from cdf_t_solver.calculators.radial_distribution_function.registry import CLOSURE_REGISTRY

def my_closure(*, r, gamma, u, sigma_ab=None):
    return np.exp(-u + gamma) - gamma - 1.0

CLOSURE_REGISTRY['MYCLOSURE'] = my_closure
system['system']['rdf']['closure']['aa'] = 'MYCLOSURE'
```

String entries are converted to uppercase by the dispatcher, so a registry key MYCLOSURE can be selected by "myclosure" or "MYCLOSURE" in the runtime dictionary.

No dedicated helper in the current snapshot — Unlike the potential registry, the closure package does not expose a register_closure(...) helper. The current extension mechanism is direct assignment into CLOSURE_REGISTRY or direct use of a callable.

29.5 Mixed closure matrix in a multicomponent system

Closures can be mixed by pair. A binary mixture can therefore use one analytical closure for aa, another for bb, and a manually defined callable for ab.

```
closure = system['system']['rdf']['closure']
closure['aa'] = 'HNC'
closure['ab'] = my_hnc      # callable
closure['bb'] = 'PY'
```

The same pair mapping is used in rdf_radial, c_analysis, and each thermodynamic state solved during boltzmann_inversion_advance. Thus, if a custom ab closure is injected before a multistate inversion, it is applied consistently to ab at every supplied state point.

29.6 Using sigma_ab inside a manual closure

The dispatcher also passes the effective pair diameter. A new closure can therefore distinguish core and non-core regions. The following is only an interface example, not a recommended physical closure:

```
def sigma_aware_example(*, r, gamma, u, sigma_ab=None):
    c = np.exp(-u + gamma) - gamma - 1.0
    if sigma_ab is not None and sigma_ab > 0.0:
        core = r < sigma_ab
```

```
# Example numerical replacement inside the core only.
c[core] = -(1.0 + gamma[core])
return c
```

29.7 Validation checklist for a newly implemented closure

- Return a finite array with exactly the same length as r and gamma.
- Protect exponentials/logarithms if the closure can overflow for a strongly repulsive potential.
- Check the dilute or weak-interaction limit against the expected theory.
- Use one physical pair entry and allow the driver to mirror it to the symmetric matrix element.
- Begin with a low-resolution test and inspect OZ convergence before using a large n_points/max_iteration.
- When comparing closures, keep potential, beta, density, grid, and tolerances unchanged.

30. Supplying a Pair Potential from a File

The most mature external-potential path in the current source is to define an interaction with filename rather than type. This path is used directly by several bundled EOS and structural examples.

30.1 Minimal input syntax

```
species = a, b
interactions primary: aa: filename = pair_aa.txt
interactions primary: ab: filename = pair_ab.txt
interactions primary: bb: filename = pair_bb.txt
```

30.2 Required numerical table

Typical two-column potential table

```
# r          U(r)
0.50000000  100.00000000
0.51000000  75.00000000
0.52000000  54.00000000
...
5.00000000  0.00000000
```

Property	Behavior in raw/hc/mf splitters
Minimum columns	at least two; first is r, second is U(r)
Additional columns	ignored by the pair-potential splitter code
Interpolation	linear
r below table	uses the first tabulated U value in the active file-backed interpolators
r above table	filled with 0.0
Units	not converted; must be consistent with beta and the chosen length/density convention
File location	resolved from the process current working directory, not automatically from ctx.input_file

30.3 Portable workflow for relative files

```
from pathlib import Path
import os

input_path = Path('system.in').expanduser().resolve()
os.chdir(input_path.parent)
```

```
# Now any filename = pair_aa.txt in system.in is resolved relative to
# the directory containing system.in.
```

The directory name is arbitrary. The important point is to establish a predictable current working directory before `hard_core_potentials`, `meanfield_potentials`, or `raw_potentials` read file-backed interactions.

30.4 Combining a file potential with analytical contributions

The splitters explicitly recognize primary, secondary, and tertiary levels. These levels are summed for the same physical pair, which allows a file-backed base interaction to be supplemented by one or two analytical terms.

File + analytic interaction composition

```
interactions primary: aa: filename = base_aa.txt
interactions secondary: aa: type = gs, sigma = 1.4, epsilon = 0.25, cutoff = 5.0
interactions tertiary: aa: type = zero

interactions primary: ab: type = mie, sigma = 1.0, epsilon = 0.5, m = 24, n = 12, cutoff = 3.5
interactions primary: bb: filename = measured_bb.txt
```

Use interaction levels for additive terms — Repeating the same primary/pair line is not a safe substitute for multiple contributions because the hierarchical parser merges dictionaries. Use primary, secondary, and tertiary for additive pieces.

30.5 What happens to a supplied file in each potential generator

Module	Role	File-backed behavior
<code>raw_potentials</code>	physical unsplit pair potential	interpolates and sums every interaction level
<code>hard_core_potentials</code>	effective core/reference analysis	loads the table, probes the repulsive region, and may infer an effective sigma through the current hard-core/BH logic
<code>meanfield_potentials</code>	mean-field representation	loads/interpolates file data and applies the module's hard-core/WCA heuristic when a very large short-range repulsion is detected
<code>total_potentials</code>	reference + MF reconstruction	combines the HC representation where <code>flag==1</code> with the mean-field representation, interpolating grid mismatches

31. Defining a New Analytical Potential Outside the Package

A user does not have to edit `pair_potential_isotropic_default.py` to add a new analytical interaction. The potential registry maps a string type to a factory that receives the parsed interaction dictionary and returns $V(r)$.

31.1 Exact potential-factory contract

```
def my_potential_factory(parameter_dict):
    # Extract any parameters defined by the user.
    ...
```

```
def V(r):
    r = np.asarray(r, dtype=float)
    return ...
return V
```

31.2 Runtime registration example

Register a custom potential before preprocessing

```
import numpy as np
from cdft_solver.generators.potential import register_isotropic_pair_potential

def screened_exponential_factory(p):
    amplitude = float(p.get('amplitude', 1.0))
    xi = float(p.get('xi', 1.0))
    cutoff = float(p.get('cutoff', 10.0))

    def V(r):
        r = np.asarray(r, dtype=float)
        out = amplitude * np.exp(-r / xi)
        out[r > cutoff] = 0.0
        return out

    return V

register_isotropic_pair_potential(
    'screened_exponential',
    screened_exponential_factory,
)
```

The type name is lowercased by the registry. The parser preserves arbitrary attributes, so an input line can use parameter names such as `amplitude` and `xi` even though the built-in package does not know them.

Input line for the registered potential

```
interactions primary: aa: type = screened_exponential, amplitude = 2.0, xi = 1.5, cutoff = 8.0
```

31.3 Complete workflow position

43. Import the registry and register the custom factory.
44. Parse the `.in` file.
45. Run `hard_core_potentials` / `meanfield_potentials` / `raw_potentials` only after registration.
46. Proceed normally to `total_potentials`, `RDF`, `free energy`, `EOS`, `coexistence`, or `external-potential generation`.

31.4 Mean-field converter registry for a custom interaction

`meanfield_potentials` first passes each analytical interaction through `POTENTIAL_CONVERTER_REGISTRY`. If no converter exists, the dictionary is copied unchanged and the same potential type is used. If a model needs a distinct perturbative/attractive representation, register a converter that rewrites the type to another registered factory.

```
from cdft_solver.generators.potential_splitter.mf_registry import register_potential_converter

def screened_to_mf(pot):
    pot['type'] = 'screened_exponential_attr'
    return pot

register_potential_converter('screened_exponential', screened_to_mf)
```

Modeling implication — Changing the mean-field converter changes which part/form of the interaction enters the perturbative free-energy and integrated-strength machinery. This must be justified physically and documented, not treated only as code plumbing.

31.5 Reusing a registered potential for an external field

`external_potential_grid` also calls `pair_potential_isotropic`. Consequently, a custom registered potential can be reused for the interaction between mobile system species and fixed external particles. The module sums $V(|\mathbf{r}-\mathbf{r}_{\text{ext}}|)$ over the positions of the external particles on the supplied spatial grid.

32. In-Memory Supplied Potential Registration

The package also contains `register_supplied_potentials` in `process_supplied_potential.py`. This is conceptually different from the simpler filename route: it receives numerical arrays already loaded into memory, creates runtime potential names, and mutates the configuration to use them.

32.1 Expected in-memory schema

Schema expected by `register_supplied_potentials`

```
supplied_potential_data = {
  'potentials': {
    'primary': {
      'aa': {
        'r': r_values,
        'U': u_values,
      },
      'ab': {
        'r': r_ab,
        'U': u_ab,
      },
    },
  }
}
```

The function searches recursively for the first potentials block. It then loops over family and pair, registers a factory named `supplied_<pair>`, and recursively changes the matching primary interaction type in `config` to that new name.

32.2 Example runtime use

```
from cdft_solver.generators.supplied_data import register_supplied_potentials

# A matching primary entry must exist so it can be replaced.
system['system']['interactions']['primary']['aa'] = {'type': 'zero'}

register_supplied_potentials(
    ctx=ctx,
    config=system,
    supplied_data=supplied_potential_data,
    export_json=True,
    export_plot=True,
)

# The primary aa type is now 'supplied_aa'.
```

32.3 Interpolation contract of `supplied_potential_factory`

Region	Value
inside tabulated interval	linear interpolation
below first r	first U value
above last r	0.0

32.4 WCA splitting intention for strongly repulsive supplied potentials

The module checks whether any supplied U exceeds 1e3. If so, it is designed to generate repulsive and attractive WCA components, register supplied_<pair>_soft and supplied_<pair>_attr, and configure the mean-field conversion to use the attractive part.

Current source bug in the hard-core branch — process_supplied_potential.py imports convert_potential_via_registry under the alias register_potential_converter, but then calls that alias with the two-argument registration signature. Therefore the hard-core supplied-potential branch is not reliable in the supplied snapshot. For a strongly repulsive tabulated potential, use interactions ... filename until this import is corrected.

The intended repair is to import register_potential_converter from mf_registry. After that correction, the design of the module is clear from the remaining code.

33. Supplied Numerical Data — Rigorous Generic Contract

process_supplied_data is a generic recursive file materializer. It does not assign physical meaning to a file; it simply replaces any nested {filename: ...} leaf by numerical x/y arrays.

33.1 Transformation performed by process_supplied_data

```
# .in file leaf
supplied_data states 1 rdf aa: filename = g_aa.txt

# Parsed dictionary leaf
{'filename': 'g_aa.txt'}

# After process_supplied_data
{
  'x': np.ndarray(...),
  'y': np.ndarray(...),
}
```

Any additional keys in that leaf are retained. The recursion also descends through lists.

33.2 Multi-column numerical files

Input table	x	y
2 columns	first column, shape (M,)	second column, shape (M,)
3+ columns	first column, shape (M,)	all remaining columns, shape (M,K)
1 column	invalid	invalid

Current exception issue — The module intends to raise SuppliedDataError for malformed/missing data, but SuppliedDataError is not defined or imported in process_supplied_data.py. Valid files work; malformed input can reveal a NameError rather than the intended custom exception.

33.3 Relative-path behavior

The loader uses ctx.work_dir only if such an attribute exists; the current ExecutionContext dataclass does not define work_dir. It therefore normally falls back to Path.cwd(). This is another reason the portable workflows change into the input-file directory before loading supplied data.

33.4 Consumer compatibility matrix

Consumer	Pair leaf expected	State metadata	Directly compatible with process_supplied_data?
boltzmann_inversion_advance	x/y or r/g	densities required; beta or temperature	YES
rdf_radial projection	r/g	densities supplied separately to rdf_radial	NO: convert x/y -> r/g
register_supplied_potentials	r/U inside potentials/family/pair	not state-based	NO: convert x/y -> r/U
c_analysis	supplied_data argument exists	densities argument required	argument is not used in the main audited source path
evaluate_canonical_state / strength kernel	supplied_data is accepted at outer level	state densities from configuration	current rdf/delta_c kernel branches explicitly invoke their structural functions with supplied_data=None

33.5 Adapter from generic x/y to rdf_radial r/g

```
def materialized_rdf_to_rg(materialized, state='1'):
    state_data = materialized['states'][state]
    result = {'rdf': {}}
    for pair, entry in state_data['rdf'].items():
        result['rdf'][pair] = {
            'r': np.asarray(entry['x'], dtype=float),
            'g': np.asarray(entry['y'], dtype=float),
        }
    return result

materialized = process_supplied_data(ctx=ctx, config=system)
rdf_constraints = materialized_rdf_to_rg(materialized, state='1')
```

33.6 Adapter from generic x/y to register_supplied_potentials r/U

```
leaf = materialized['potentials']['primary']['aa']
supplied_for_registry = {
    'potentials': {
        'primary': {
            'aa': {
                'r': np.asarray(leaf['x'], dtype=float),
                'U': np.asarray(leaf['y'], dtype=float),
            }
        }
    }
}
```

For ordinary tabulated pair potentials, the direct filename interaction is usually simpler and more robust than this conversion route.

34. Combining Supplied and Computed RDF Data in rdf_radial

rdf_radial contains an advanced projection mode that is separate from potential inversion. It allows selected $g_{ij}(r)$ to be treated as externally fixed structural information while the remaining pair correlations are determined self-consistently from OZ and the chosen closures.

34.1 Internal projection sequence

47. Run an ordinary unconstrained OZ calculation for all pairs.
48. Interpolate each supplied RDF pair onto the internal radial grid and mark that matrix element fixed.
49. Replace $h_{ij}=g_{ij}-1$ for the fixed pairs by the supplied values.

50. Transform the complete h matrix to k space.
51. Use $C(k)=H(k)[I+\rho H(k)]^{-1}$ to obtain a projected direct-correlation matrix.
52. Inverse-transform $c(k)$ back to $c(r)$.
53. Freeze c for the supplied pairs and allow closure updates only for the free pairs.
54. Resolve OZ and iterate the projection until the free-pair h functions stop changing within tolerance.

34.2 Example: supply aa, compute ab and bb

```

rdf_constraints = {
    'rdf': {
        'aa': {
            'r': r_sim,
            'g': g_aa_sim,
        }
    }
}

rdf_out = rdf_radial(
    ctx=ctx,
    rdf_config=system,
    densities=np.array([rho_a, rho_b], dtype=float),
    supplied_data=rdf_constraints,
    export=True,
    plot=True,
    filename_prefix='hybrid_structure',
)

```

The potentials are still the interactions defined in the configuration. This operation constrains structure; it does not update or infer a potential.

34.3 Supplying more than one pair

Any subset can in principle be fixed: aa only, ab only, aa+ab, or all pair RDFs. The free pairs are those for which `fixed_mask` is False.

Pair-key asymmetry in this consumer — `rdf_radial` process_supplied_rdf checks exact keys as it loops over ordered matrix positions and does not perform the AB/BA fallback and symmetric assignment used by the advanced inversion processor. For robust usage, supply canonical pair keys consistent with species order; if both matrix directions must be fixed explicitly, provide both.

35. Multistate Effective-Potential Inversion in Detail

`boltzmann_inversion_advance` accepts one or many target RDF state points and updates one common pair-potential matrix. It is therefore capable of both ordinary single-state inversion and a more restrictive multistate coarse-graining problem.

35.1 Accepted state container names

`process_supplied_rdf_multistate` looks for a top-level states mapping, then state, then falls back to treating the `supplied_data` object itself as the mapping of state names to state dictionaries.

35.2 State density formats

Input	Example	Result for species [a,b]
dict by species	<code>{"a":0.10,"b":0.02}</code>	[0.10, 0.02]
dict missing b	<code>{"a":0.10}</code>	[0.10, 0.00]

scalar	0.10	[0.10, 0.10]
length-1 array	[0.10]	[0.10, 0.10]
length-2 array	[0.10,0.02]	[0.10,0.02]

A multidimensional density array or an array with the wrong length raises a ValueError.

35.3 Beta and temperature metadata

If beta is present, it is used directly. Otherwise, if temperature is present, $\beta = 1/\text{temperature}$. If neither appears, beta defaults to 1.0. The first state beta becomes `beta_ref` for the common internal potential representation.

35.4 RDF leaves accepted by inversion

For each pair the inversion processor accepts either x/y or r/g. This is why `process_supplied_data` output can be passed directly to `boltzmann_inversion_advance`.

The supplied RDF is linearly interpolated onto the inversion grid. Below the first r point the first g value is used; above the last supplied r it approaches 1.0. The interpolated curve is then passed through `clean_gr_tail` before being stored in `g_target`.

35.5 Missing pair RDFs are allowed

The processor allocates `g_target` as zeros and `fixed_mask` as False, then sets only those pairs that are actually found in the supplied state. Missing pair data are therefore not invented. This is what permits known model interactions and unknown/inverted interactions to coexist.

35.6 Partial inversion: known aa and bb, infer ab only

Mixed known + inverted pair workflow

```
species = a, b

# Known pair interactions
interactions primary: aa: filename = known_aa.txt
interactions primary: bb: type = gs, sigma = 1.414, epsilon = 2.0, cutoff = 5.0
# No ab model interaction is required if ab is to be inferred.

rdf closure aa = hnc
rdf closure ab = py
rdf closure bb = hnc
rdf alpha_max = 0.1
rdf max_iteration = 50000
rdf max_iteration_ibi = 5000
rdf alpha_ibi_max = 0.1
rdf tolerance = 0.00001
rdf ibi_tolerance = 0.0001
rdf r_max = 15.0
rdf n_points = 1500

supplied_data states 1 rdf ab: filename = g_ab_state1.txt
supplied_data states 1 densities a = 0.10
supplied_data states 1 densities b = 0.02
supplied_data states 1 beta = 1.0
```

35.7 How the inversion mask is actually created

55. `u_matrix` starts at zero and `invert_mask` starts True for every pair.
56. Any preprocessed model pair potential is interpolated into `u_matrix`. A nonzero pair is initially marked `invert_mask=False`.

57. For each supplied target RDF pair, the first-state g is converted to $-\ln(\max(g, g_{\text{floor}}))$ and written into `u_matrix`, σ is estimated from the RDF core, and `invert_mask` is set True.
58. During IBI, a correction is accumulated only where `fixed_mask` is True. Therefore an unsupplied known model pair receives no IBI update.
59. A pair with neither a model potential nor supplied RDF remains effectively zero because no correction is accumulated.

If model potential and RDF are both supplied for the same pair — The current initialization replaces that pair by $-\ln(g)$ and marks it for inversion. The model potential is not retained as the IBI starting guess for a supplied pair.

35.8 Multiple thermodynamic state points constraining one potential

Three-state inversion constraint

```
supplied_data states 1 rdf ab: filename = g_ab_rho1_T1.txt
supplied_data states 1 densities a = 0.10
supplied_data states 1 densities b = 0.02
supplied_data states 1 beta = 1.0

supplied_data states 2 rdf ab: filename = g_ab_rho2_T1.txt
supplied_data states 2 densities a = 0.18
supplied_data states 2 densities b = 0.04
supplied_data states 2 beta = 1.0

supplied_data states 3 rdf ab: filename = g_ab_rho2_T2.txt
supplied_data states 3 densities a = 0.18
supplied_data states 3 densities b = 0.04
supplied_data states 3 temperature = 1.25
```

The routine currently assigns equal state weights, $w_{\text{state}}=1/N_{\text{states}}$. A separate OZ problem is solved at each state on every IBI iteration, but all states update the same common `u_matrix`.

35.9 Numerical IBI correction

For state s and supplied pair ij , the implemented correction is proportional to $(\beta_{\text{ref}}/\beta_s) \ln[g_{\text{pred},s}(r)/g_{\text{target},s}(r)]$. Only radii with $g_{\text{target}} > 1e-4$ enter the logarithmic update. State corrections are averaged and then multiplied by adaptive `alpha_ibi`.

35.10 Adaptive `alpha_ibi`

Quantity	Current behavior
initial <code>alpha_ibi</code>	0.01
minimum	$1e-4$
maximum	<code>rdf alpha_ibi_max</code>
adaptation interval	every 10 IBI iterations after iteration 1
adaptation signal	ratio of previous maximum RDF error to current maximum RDF error, clipped before applying a power 0.5
convergence metric	maximum absolute $ g_{\text{pred}} - g_{\text{target}} $ over all target pairs and states
convergence target	<code>rdf ibi_tolerance</code>

35.11 Post-inversion hard-core/reference analysis

After IBI convergence, the routine continues with substantial reference analysis. It detects near-zero RDF core intervals, identifies hard-core pairs, constructs repulsive references, optimizes sigma collectively over all supplied states, evaluates Barker-Henderson diameters, generates attractive/reference decompositions, and computes several state-resolved structural diagnostics.

35.12 Main exported files from an advanced inversion

Export	Meaning
<prefix>_potential.json	compact final pair potentials: r, u_r, sigma for every matrix pair
result_sigma_analysis.json	sigma_opt, sigma_bh, reference/real potentials and state-resolved g/c/gamma packages
result_G_of_r.json	coupling-integrated G(r), G_u(r), and attractive potential information
delta_c_by_state.json	real/reference direct correlations and several Delta-c combinations
plots/	potential and RDF/reference diagnostics when enabled

The function finally returns (u_matrix / beta_ref, sigma_matrix), with shapes (N,N,Nr) and (N,N), respectively.

36. Advanced Data-Combination Recipes

36.1 Recipe A — tabulated potential + manual closure

```
# system.in
species = a
interactions primary: aa: filename = measured_u.txt
rdf closure aa = py
rdf alpha_max = 0.05
rdf max_iteration = 100000
rdf tolerance = 1e-6
rdf beta = 1.0
rdf r_max = 12.0
rdf n_points = 1200
# workflow.py
system = super_dictionary_creator(ctx=ctx, export_json=True)

def my_closure(*, r, gamma, u, sigma_ab=None):
    return np.exp(-u + gamma) - gamma - 1.0

system['system']['rdf']['closure']['aa'] = my_closure
rdf_out = rdf_radial(
    ctx=ctx,
    rdf_config=system,
    densities=np.array([0.2]),
    export=True,
)
```

Here the potential is external numerical data, while the closure is external Python logic. They enter the solver independently.

36.2 Recipe B — analytical self interactions + inverted cross interaction

60. Define aa and bb under interactions using built-in, registered, or file-backed potentials.

61. Supply `g_ab` only.
62. Materialize the RDF files.
63. Call `boltzmann_inversion_advance`.
64. Reuse the exported ab potential in a later thermodynamic calculation if desired.

36.3 Recipe C — inversion at many states, then reuse as a potential file

A transparent two-stage scientific workflow is often preferable: first infer a potential from several states and save `<prefix>_potential.json`; next export the desired pair as a two-column `r,U` text file and reference that file from a new EOS/coexistence input. The inferred potential then becomes an explicit, archivable input to the thermodynamic stage.

36.4 Recipe D — supplied RDF for one pair, closure solution for the others

When the physical potential is already known and only part of the structure is measured, use `rdf_radial` constrained projection rather than inversion. The supplied pair is imposed structurally; free pairs are reconstructed from OZ plus their closures.

36.5 Recipe E — file + analytic + custom potential in one mixture

```
# aa is read from file
interactions primary: aa: filename = aa.txt

# ab is a runtime registered functional form
interactions primary: ab: type = screened_exponential, amplitude = 1.2, xi = 0.8, cutoff = 5.0

# bb is a sum of two built-in contributions
interactions primary: bb: type = gs, sigma = 1.4, epsilon = 2.0
interactions secondary: bb: type = mie, sigma = 1.0, epsilon = 0.2, m = 24, n = 12, cutoff = 3.0
```

The potential generators do not require all pairs to be represented by the same mechanism.

36.6 Recipe F — manual closure during multistate inversion

```
system = super_dictionary_creator(ctx=ctx)
supplied = process_supplied_data(ctx=ctx, config=system)

# Use a new closure only for the cross pair.
system['system']['rdf']['closure']['ab'] = my_closure

u, sigma = boltzmann_inversion_advance(
    ctx=ctx,
    rdf_config=system,
    supplied_data=supplied,
    filename_prefix='effective_multistate',
)
```

Because the `pair_closures` matrix is built once from `rdf_config` and reused at every state, the same manual ab closure enters every state-specific OZ solve consistently.

37. Compatibility and Implementation Notes for Advanced Users

Feature	Observed source behavior	Recommended practice
Custom closure in <code>.in</code> file	plain text cannot encode a Python callable	parse first, then inject callable; keep workflow script as provenance
Custom closure registry	CLOSURE_REGISTRY is directly mutable	register before solver call; use a unique uppercase name
Custom potential	formal <code>register_isotropic_pair_potential</code> API exists	register before any potential preprocessing

Tabulated potential path	read relative to os.getcwd()	chdir to input directory before loading
Generic supplied data	creates x/y	use directly for inversion; adapt for other consumers
rdf_radial supplied RDF	expects r/g	convert materialized x/y explicitly
in-memory supplied potential	expects r/U	adapt generic data or use direct filename route
hard-core supplied-potential registrar	contains wrong converter-function alias	prefer filename route until source is repaired
c_analysis supplied_data	signature accepts it but main path ignores it	treat c_analysis as potential+density driven in this snapshot
EOS strength-kernel supplied_data	outer function accepts it, internal rdf/delta_c branches pass None	do not assume EOS reads external RDF data without source modification
multistate weights	uniform and hard-coded	document that all states have equal weight
initial guess for supplied inversion pair	set from -ln(g)	a simultaneously supplied model potential is overwritten as starting guess

37.1 Source improvements that would strengthen the public API

65. Add register_closure(name, fn, overwrite=False) to mirror the potential registry.
66. Add input_dir/work_dir to ExecutionContext and resolve all external files relative to ctx.input_file by default.
67. Standardize supplied numerical data to one canonical schema and centralize x/y <-> r/g <-> r/U adapters.
68. Fix process_supplied_potential.py to import register_potential_converter instead of aliasing convert_potential_via_registry.
69. Allow state weights to be supplied explicitly in multistate inversion.
70. Allow a supplied RDF pair to optionally retain a model potential as its IBI starting guess.
71. Mirror supplied AB/BA automatically in rdf_radial as the inversion processor already does.
72. Add preflight validation for closure callable signatures, pair completeness, state densities, and supplied data dimensions.

38. Advanced Quick Reference

Goal	Use	Input/data form
Use potential from simulation file	interactions pair filename	2-column r,U text
Add new analytical potential	register_isotropic_pair_potential	factory(dict)->V(r), then type=<registered name>
Add new closure	callable in rdf.closure or CLOSURE_REGISTRY	function(r=,gamma=,u=,sigma_ab=)->c(r)
Invert one pair only	known model pairs + supplied RDF for unknown pair	multistate supplied_data with only target pair present
Invert across several states	boltzmann_inversion_advance	states with densities, beta/temperature, pair RDFs
Fix measured RDF and compute other RDFs	rdf_radial projection	supplied_data {rdf:{pair:{r,g}}}
Reuse inverted potential later	export to r,U table and use filename	explicit staged workflow
Use custom potential for fixed external particles	external_potential_grid + same potential registry	external interaction dict uses registered type

39. Final Summary of the Extended Manual

The package is more extensible than its bundled examples alone suggest. A user can mix analytical and file-backed potentials across different species pairs, decompose one pair into primary/secondary/tertiary contributions, introduce a completely new potential factory at runtime, replace one closure by a Python callable, constrain selected RDFs while calculating the remaining structure, or infer only selected unknown interactions from RDF data collected at several state points.

The most important practical rule is to track the data contract at every boundary: `.in` text -> parsed dictionary; `filename` -> numerical interpolation; generic supplied file -> `x/y`; structural projection -> `r/g`; supplied-potential registration -> `r/U`; RDF state -> `g_target/fixed_mask`; and inversion -> a common `u_matrix` plus `sigma_matrix`. When those boundaries are documented explicitly, the same `cDFT_solver` modules can be combined in many different but reproducible scientific workflows.